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Kim et al.

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(54) **SEMICONDUCTOR DEVICES**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 65 days.

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(22) Filed: **Jul. 11, 2023**

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Related U.S. Application Data

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(30) **Foreign Application Priority Data**

Dec. 15, 2020 (KR) 10-2020-0175579

(51) **Int. Cl.**

H10B 12/00 (2023.01)
H10D 84/85 (2025.01)

(52) **U.S. Cl.**

CPC ***H10B 12/50*** (2023.02); ***H10B 12/315*** (2023.02); ***H10D 84/85*** (2025.01)

(58) **Field of Classification Search**

CPC H10B 12/50; H01L 27/092
See application file for complete search history.

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(74) *Attorney, Agent, or Firm* — Muir Patent Law, PLLC

(57) **ABSTRACT**

A semiconductor device includes first and second trenches in respective first and second regions in a substrate, a first isolation structure having a first inner wall oxide pattern, a first liner, and a first filling insulation pattern sequentially stacked I the first trench, a second isolation structure having a second inner wall oxide pattern, a second liner, and a second filling insulation pattern sequentially stacked I the second trench, a first gate structure having a first high-k dielectric pattern, a first P-type metal pattern, and a first N-type metal pattern sequentially stacked on the first region, and a second gate structure having a second high-k dielectric pattern and a second N-type metal pattern sequentially stacked on the second region, wherein the first and second liners protrude above upper surfaces of the first and second inner wall oxide patterns and the first and second filling insulation patterns, respectively.

20 Claims, 25 Drawing Sheets

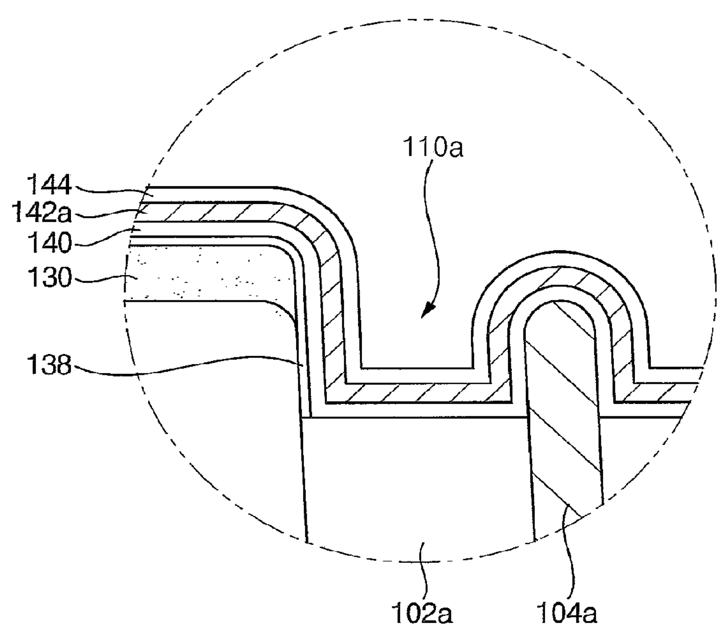
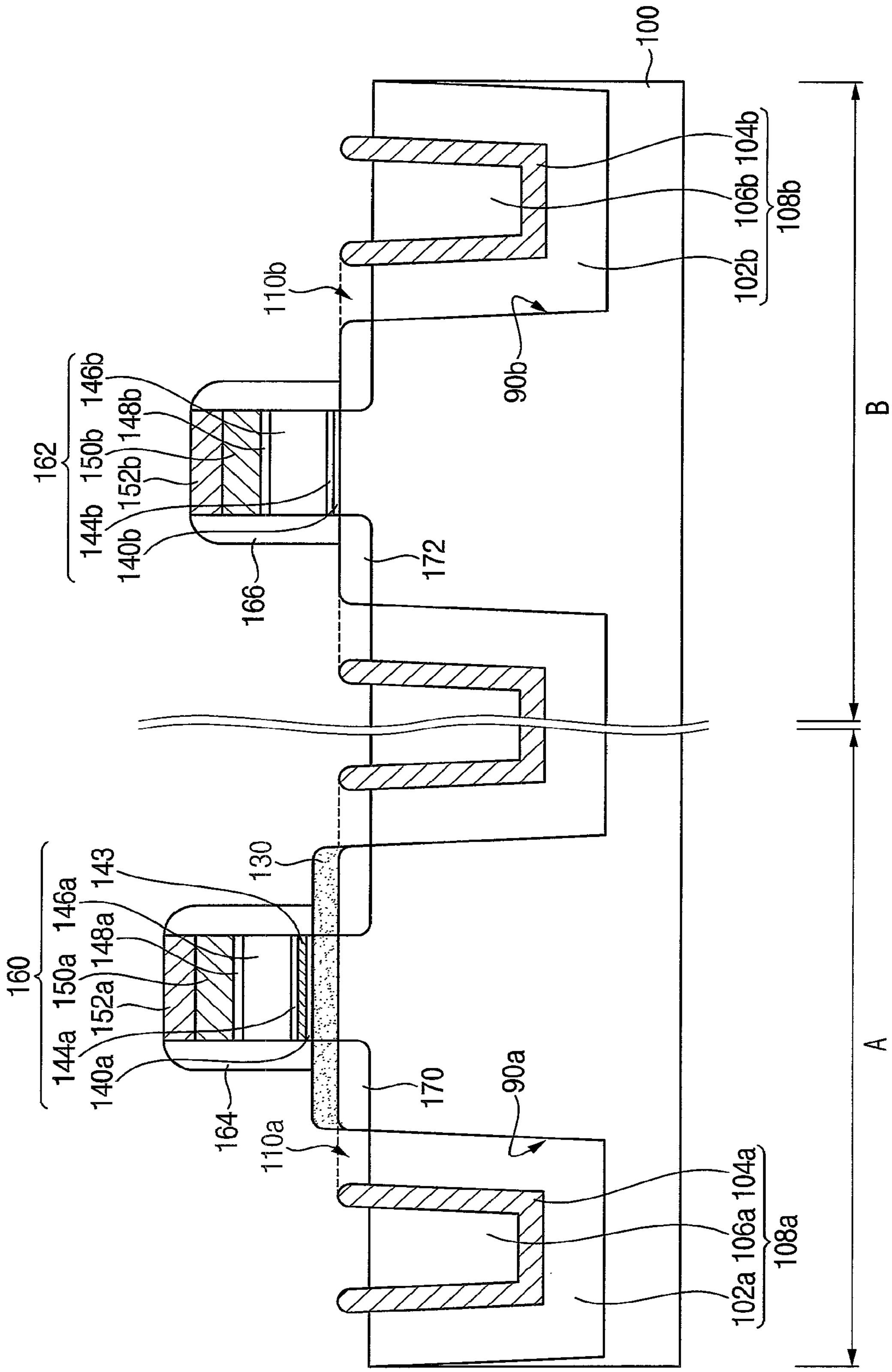


FIG. 1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140a
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FIG. 2

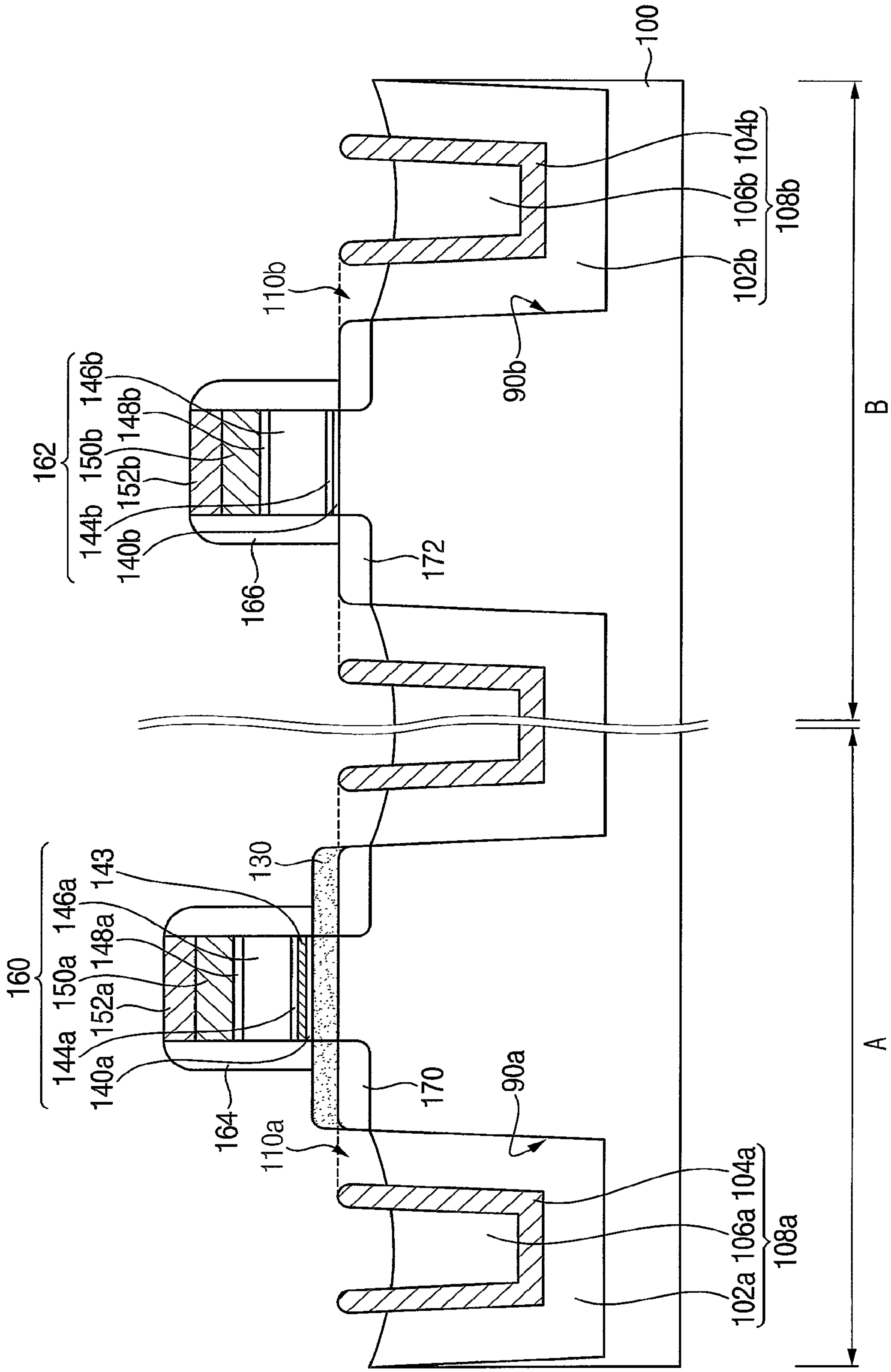
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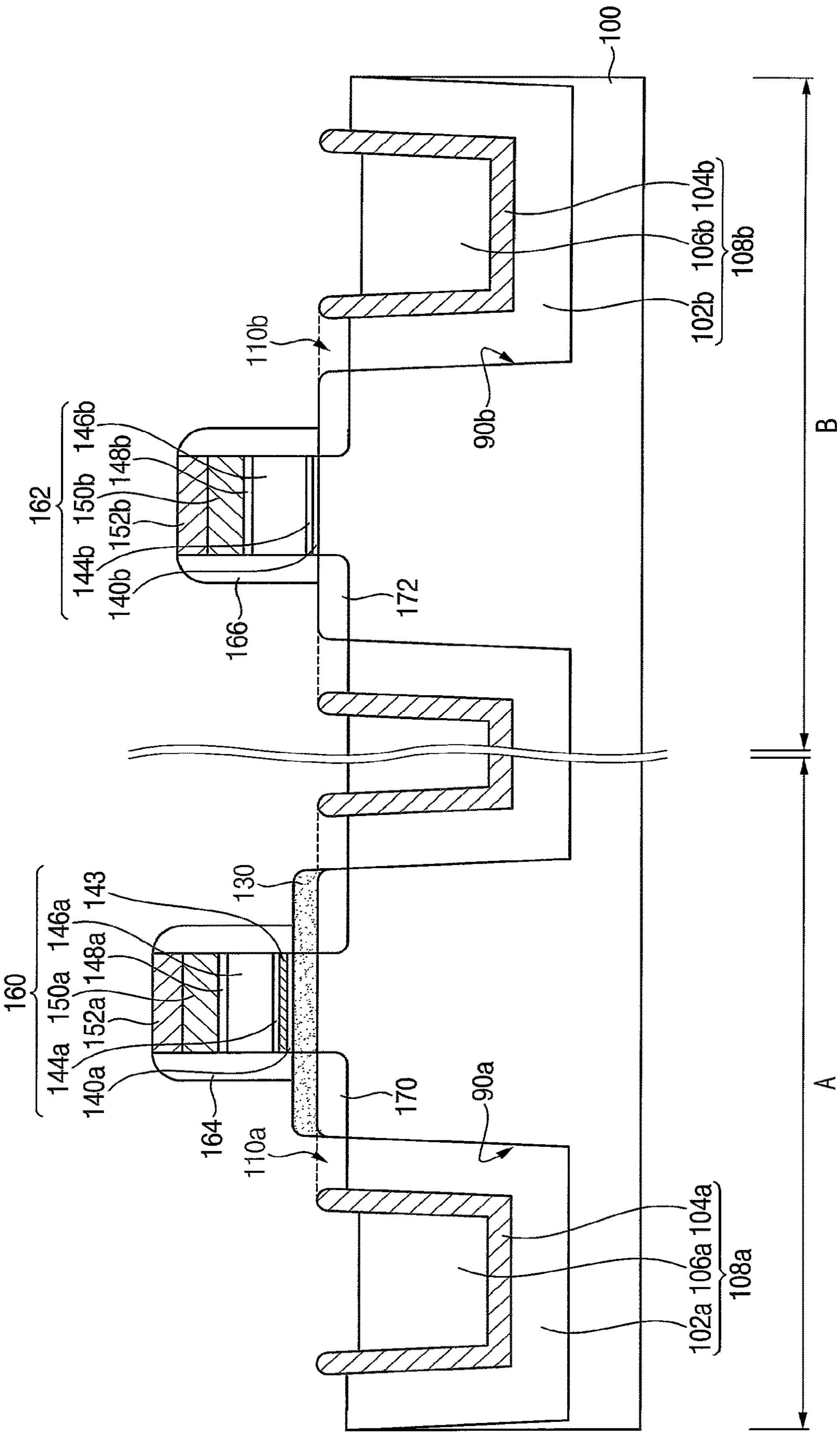
FIG. 3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146a
143
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108a
102a

FIG. 4

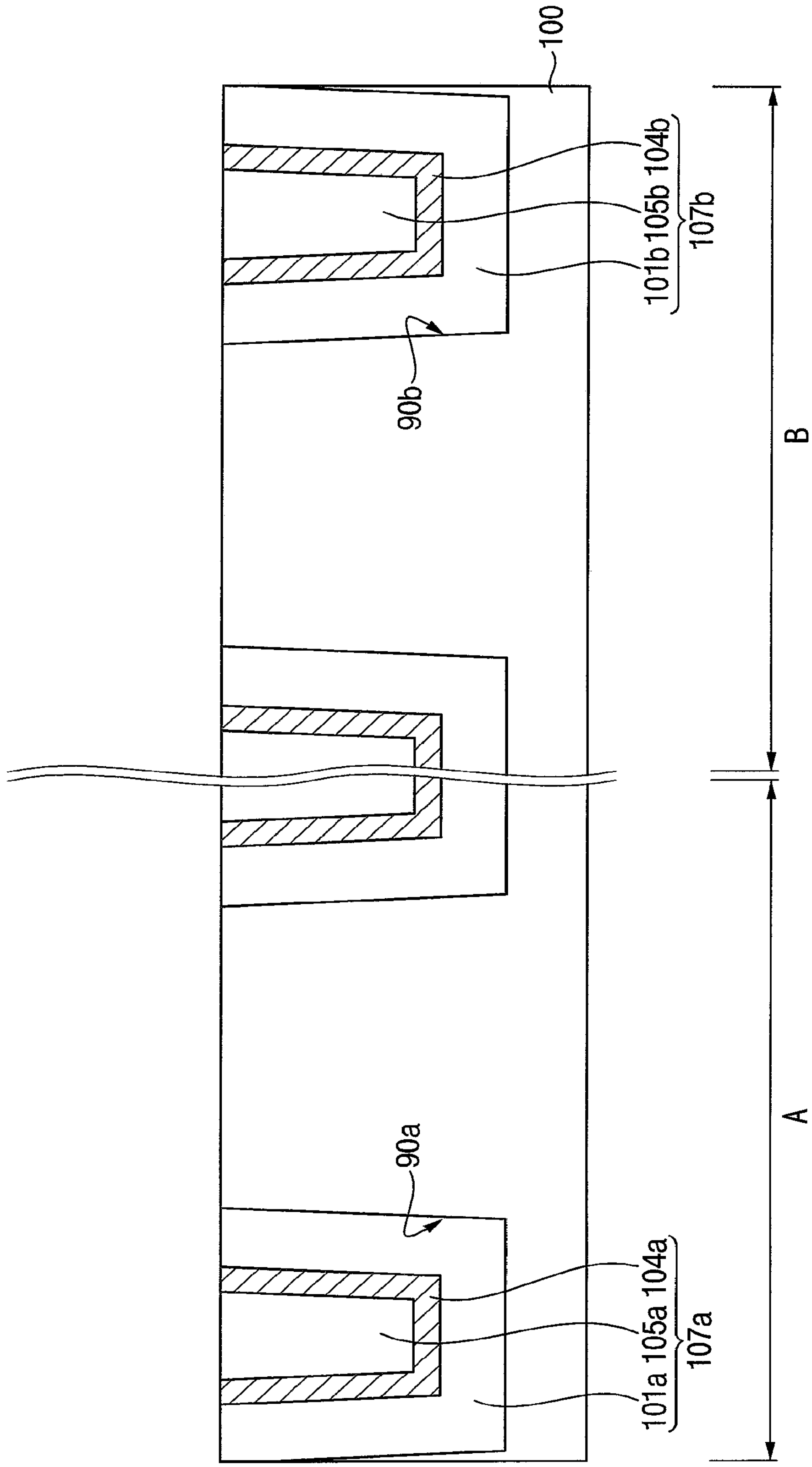
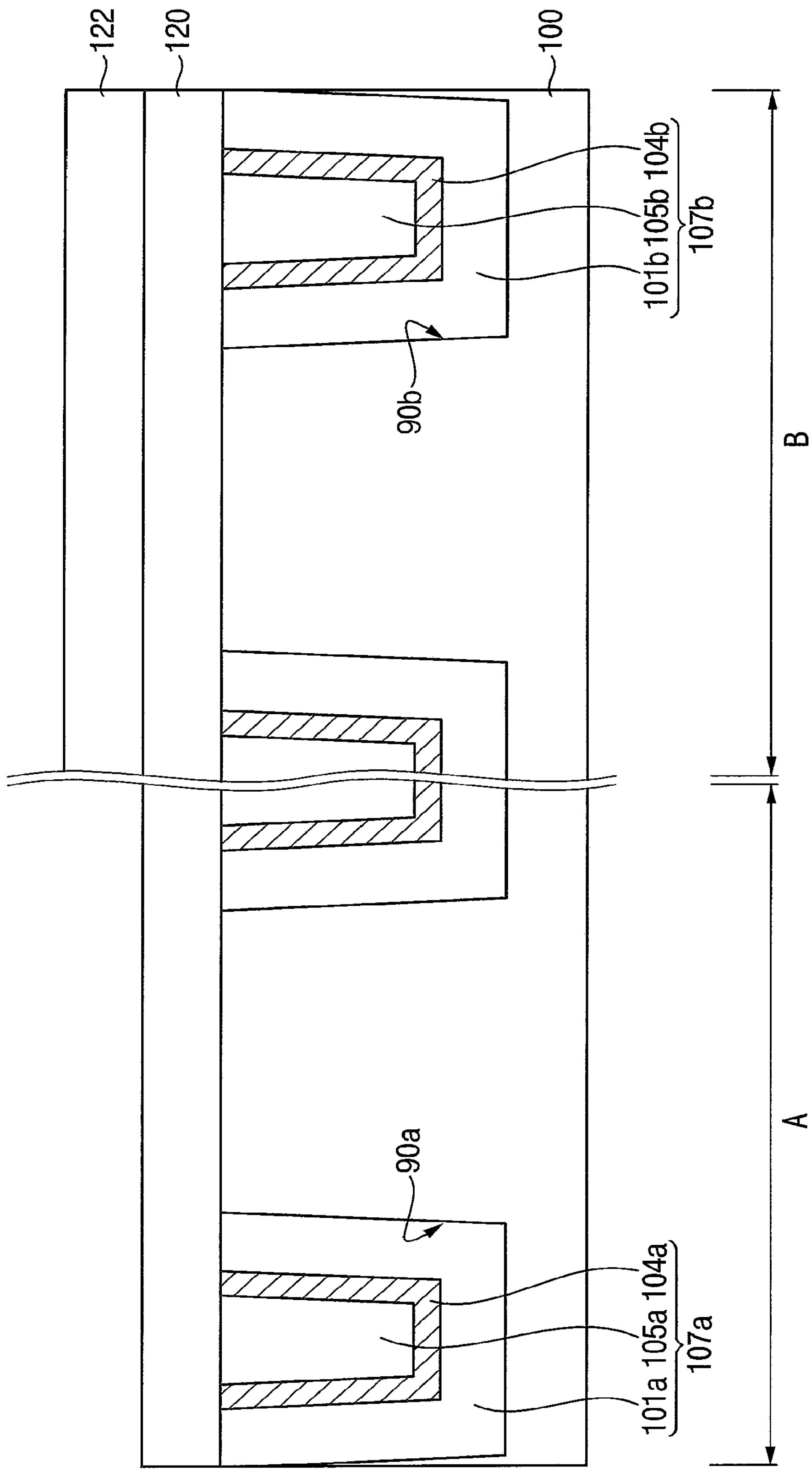


FIG. 5



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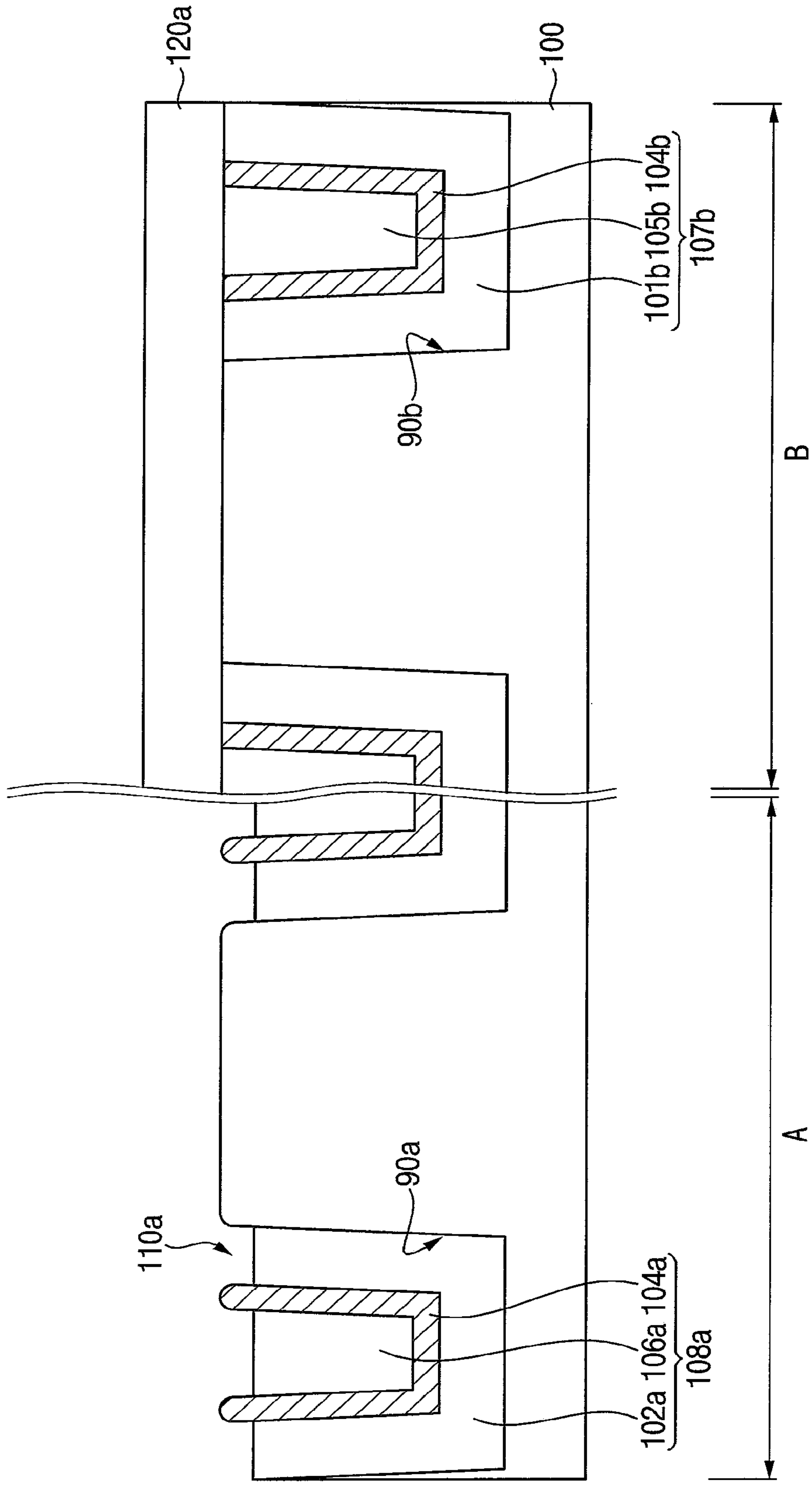
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FIG. 7

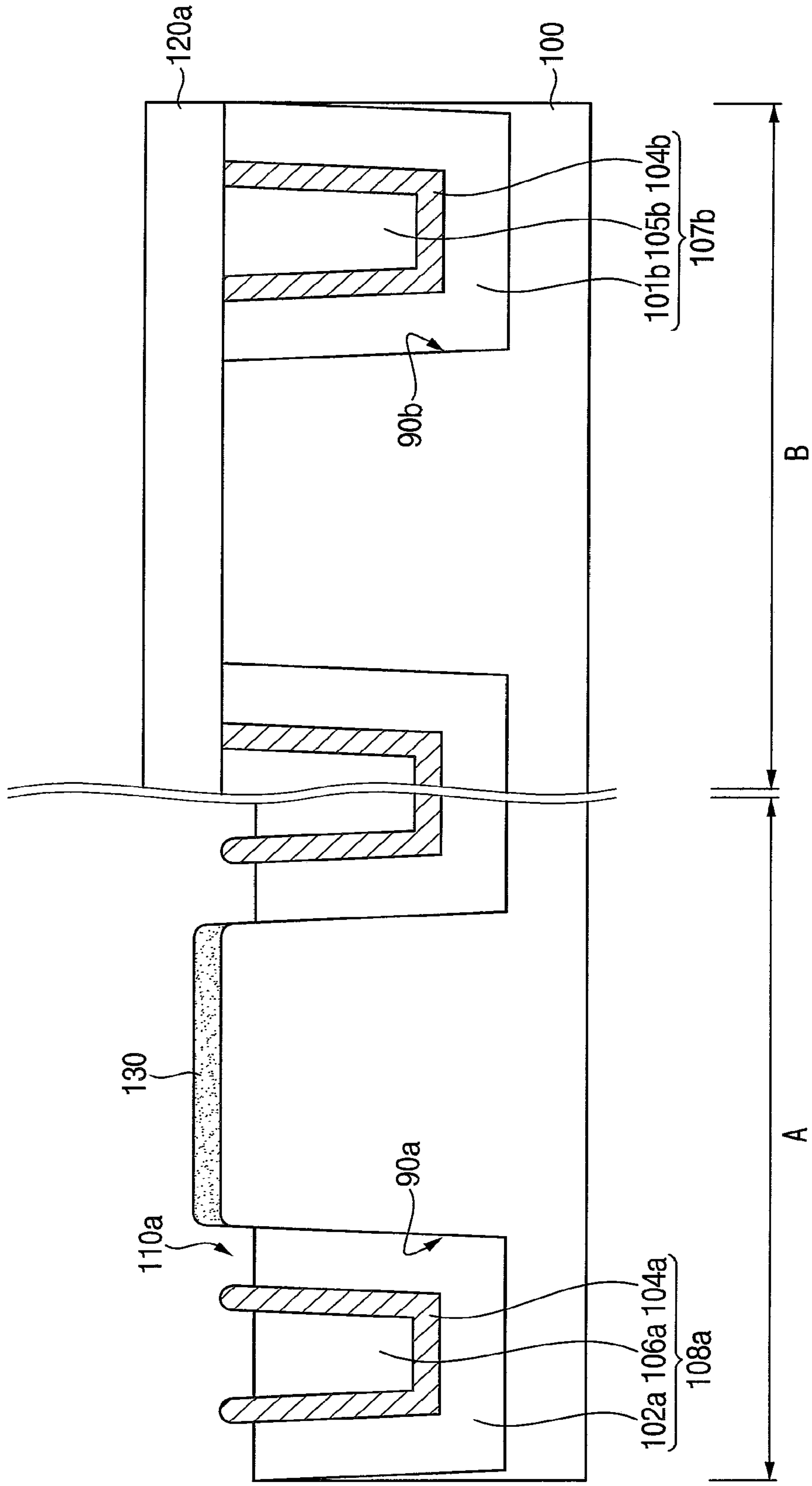
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FIG. 8

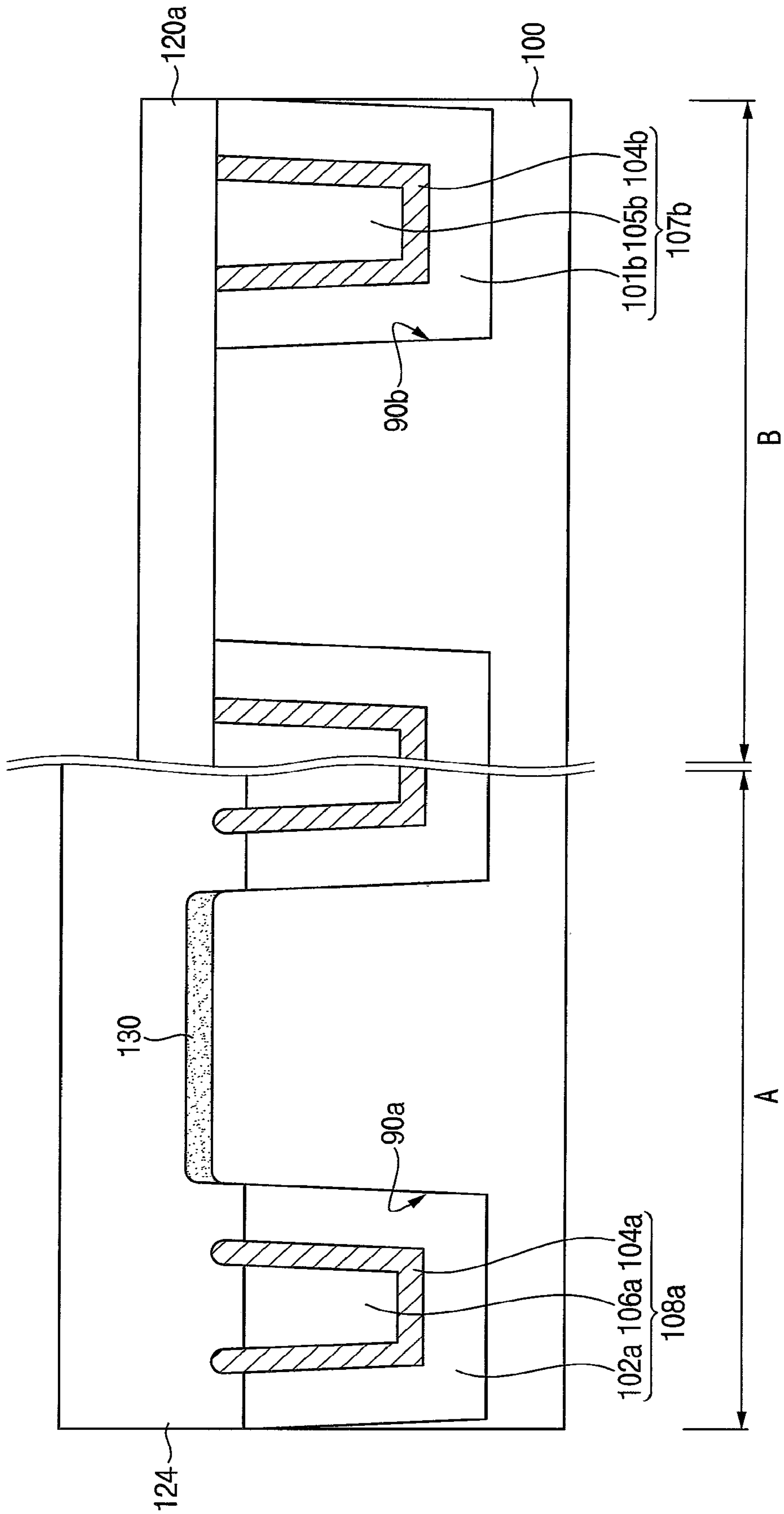
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FIG. 9

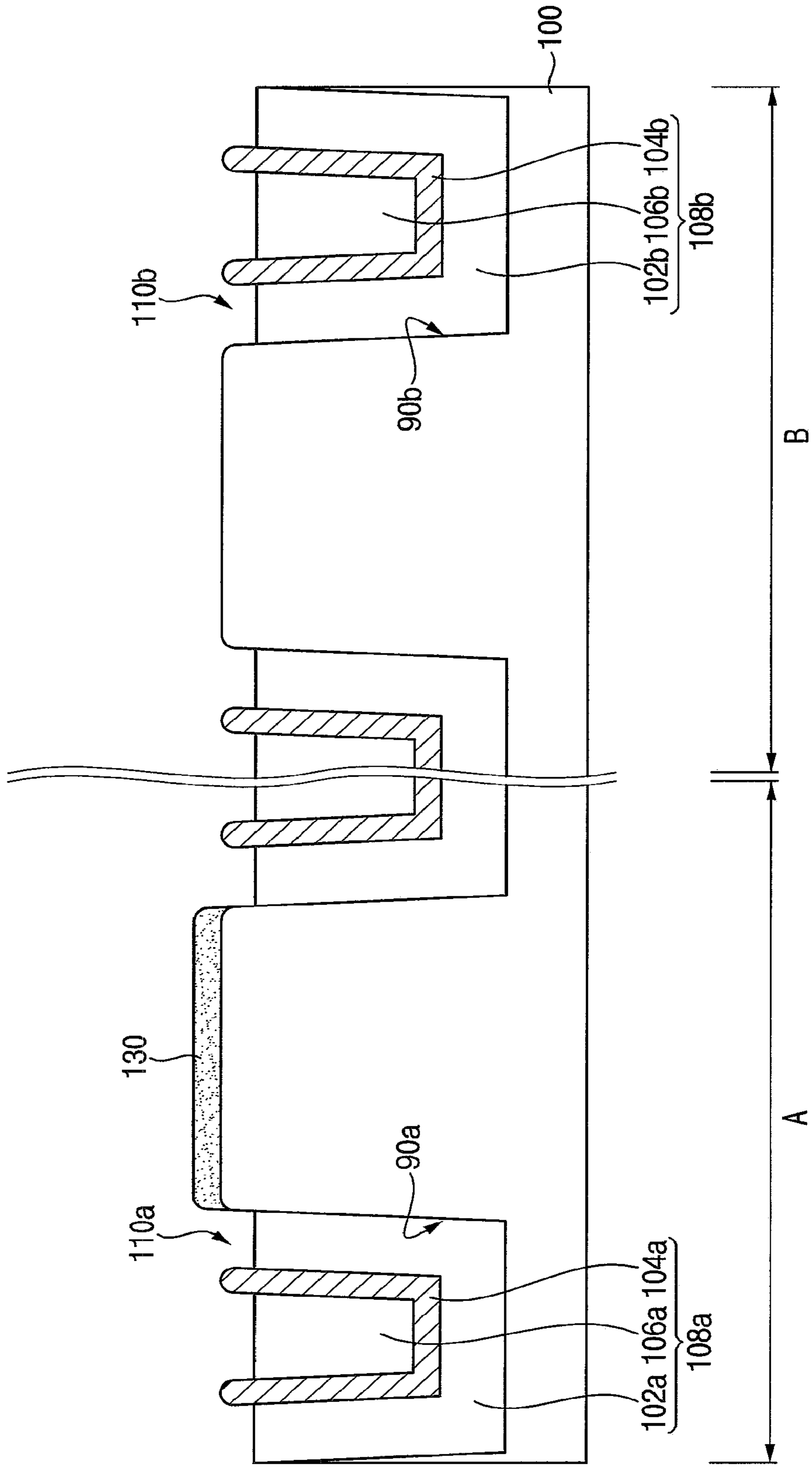
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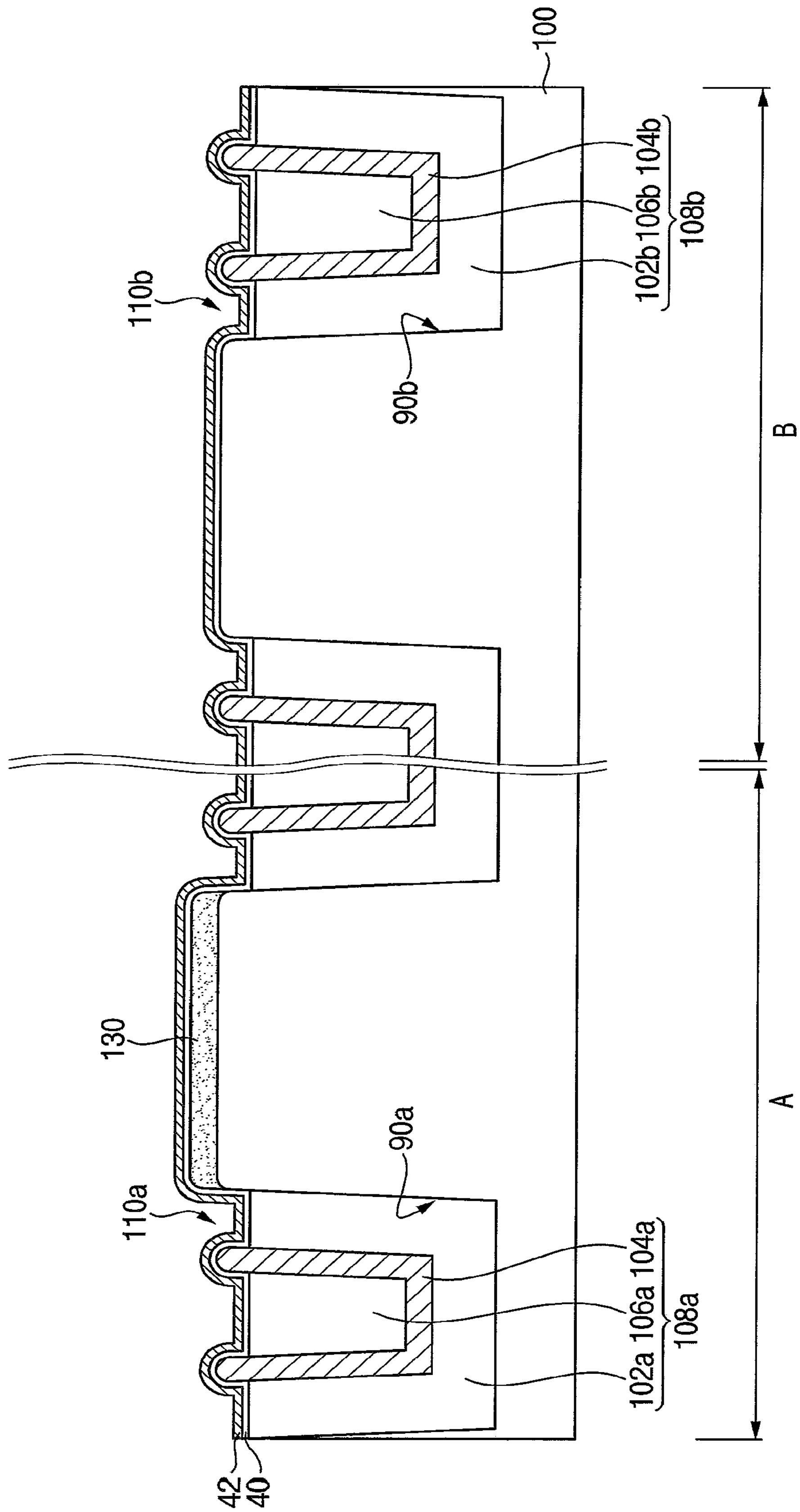
FIG. 10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108a
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140

FIG. 11

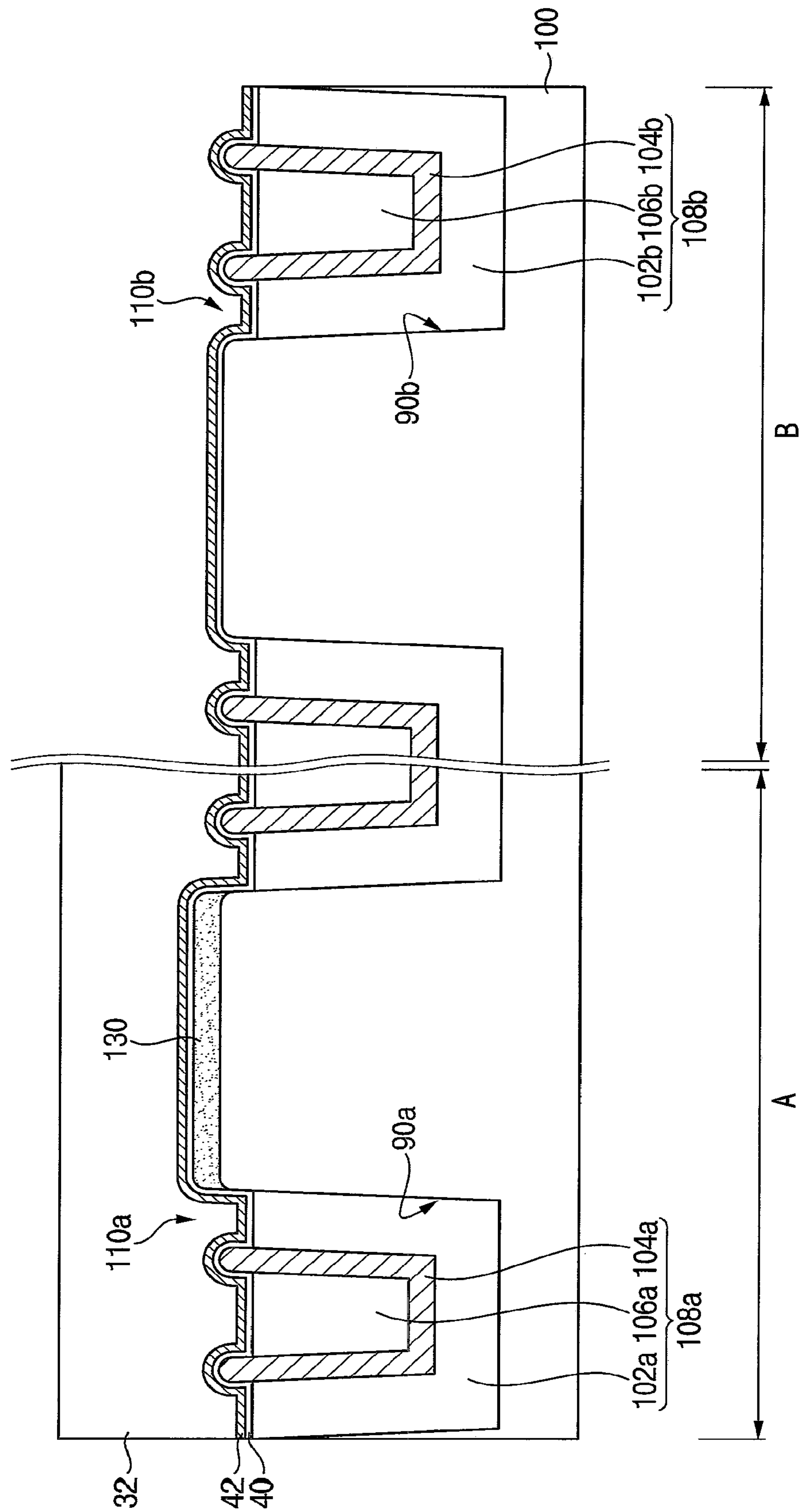
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FIG. 12

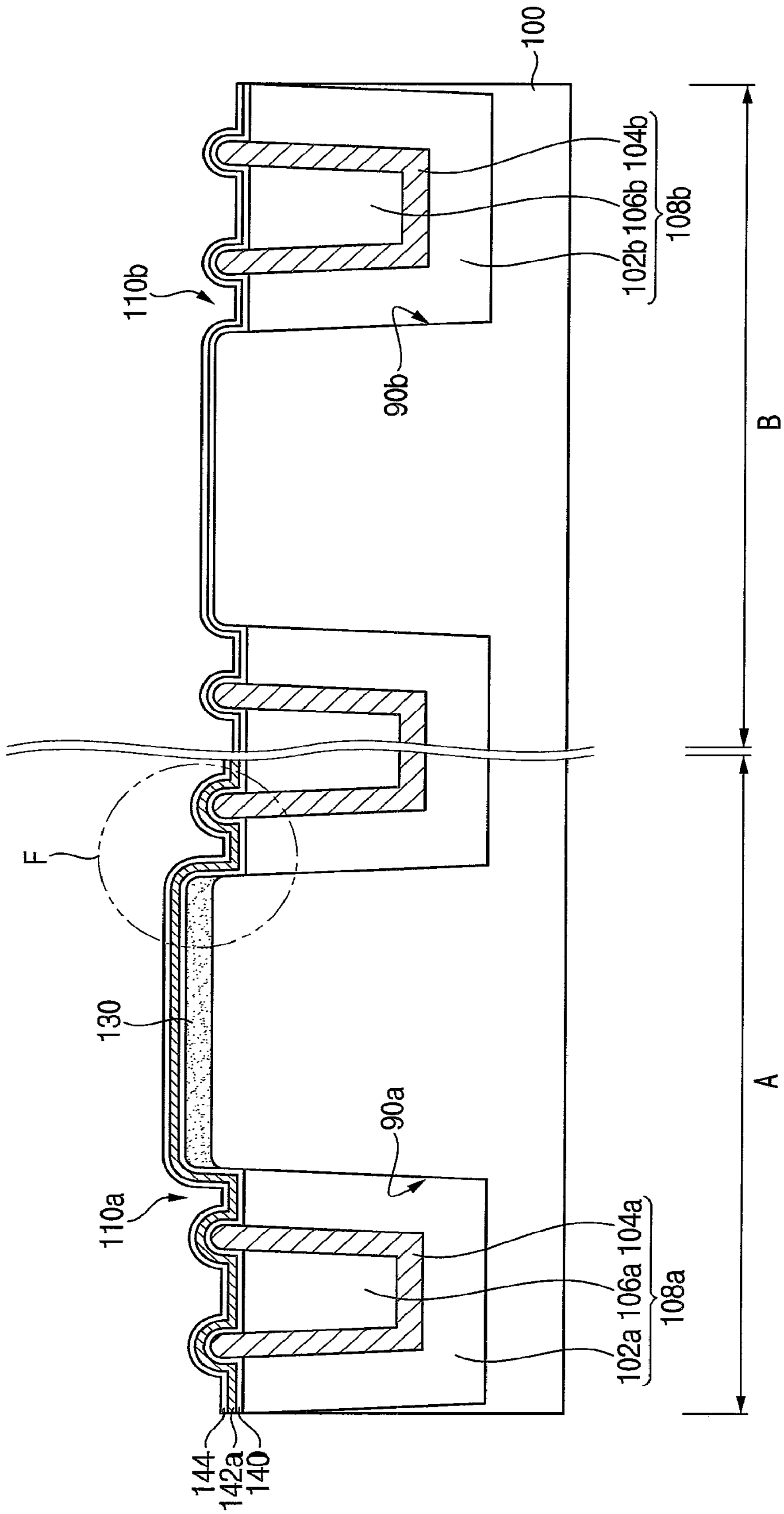
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FIG. 13

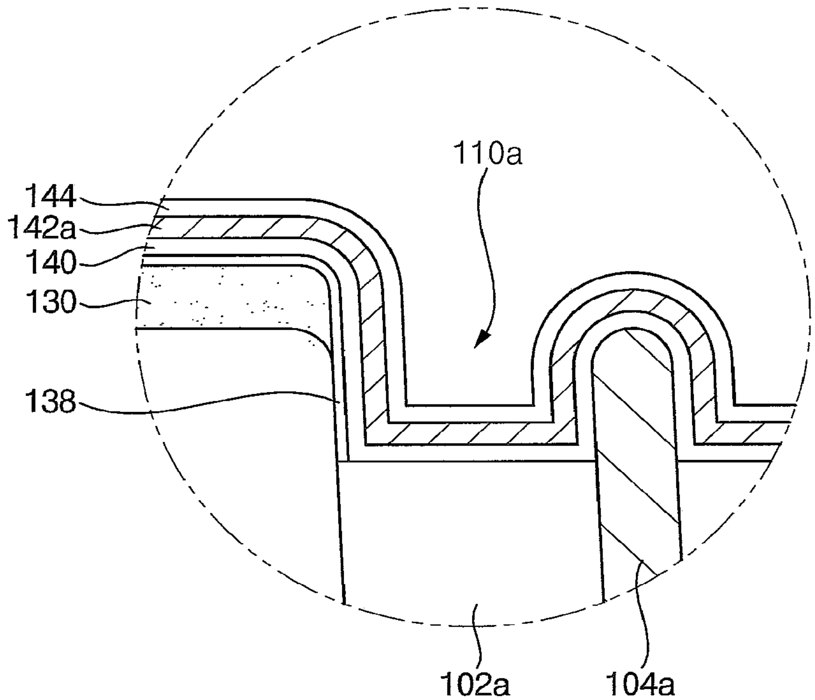
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FIG. 14

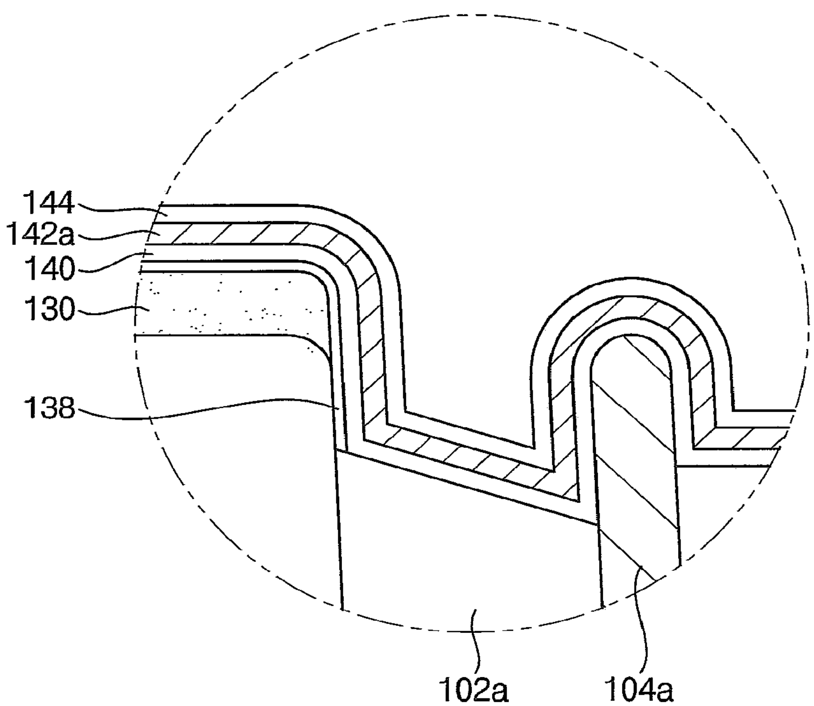
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FIG. 15

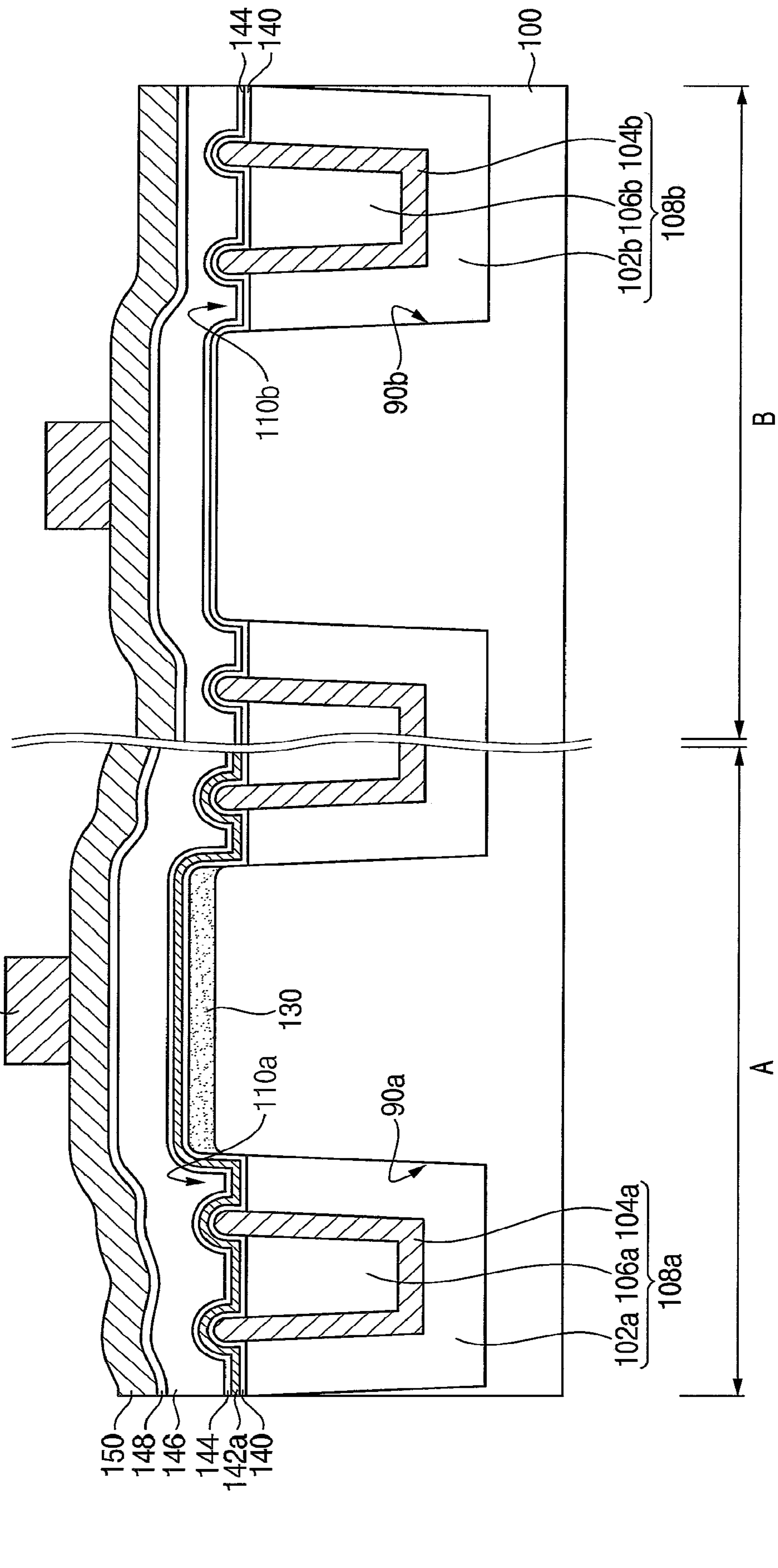
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FIG. 16

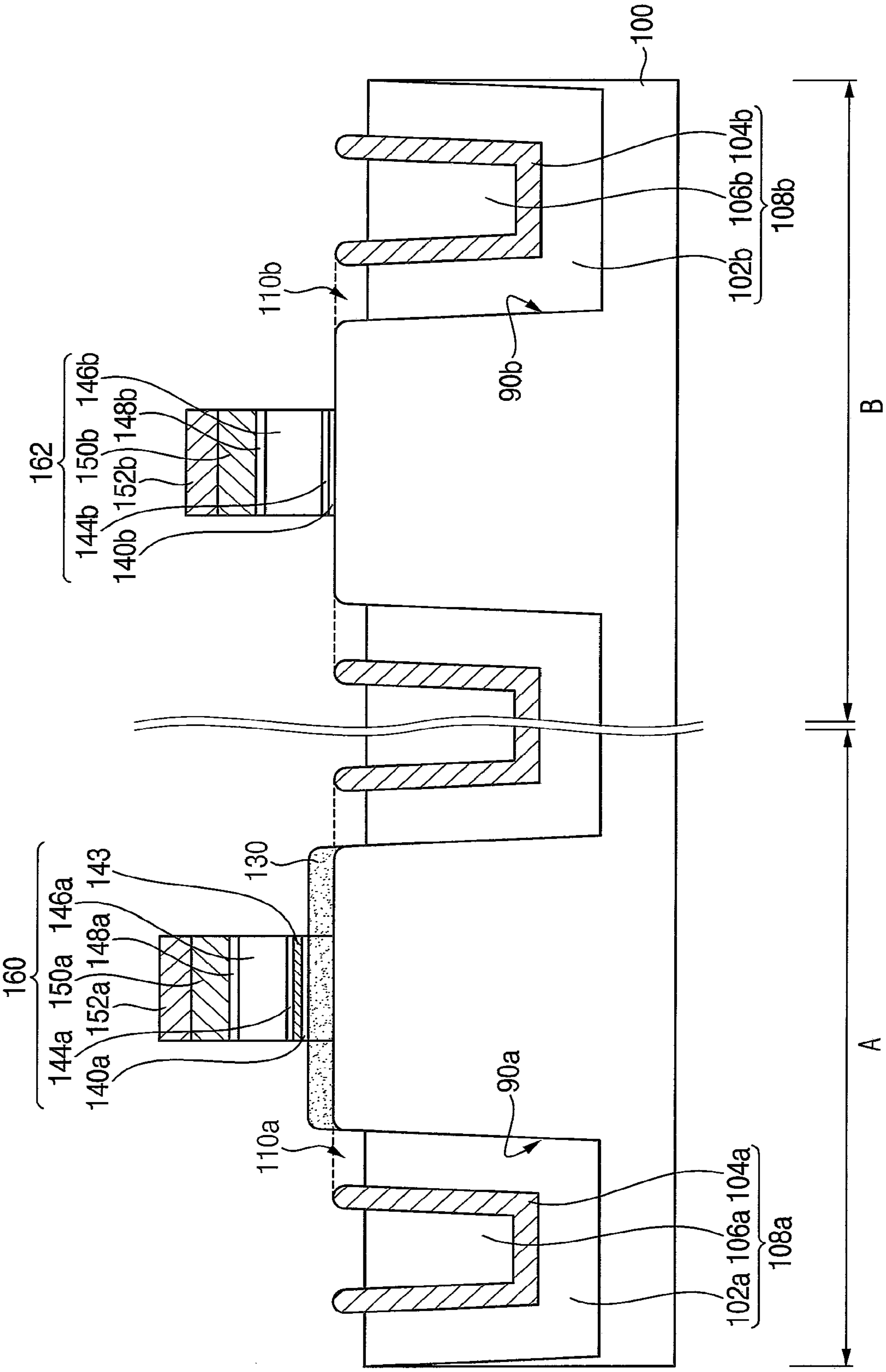
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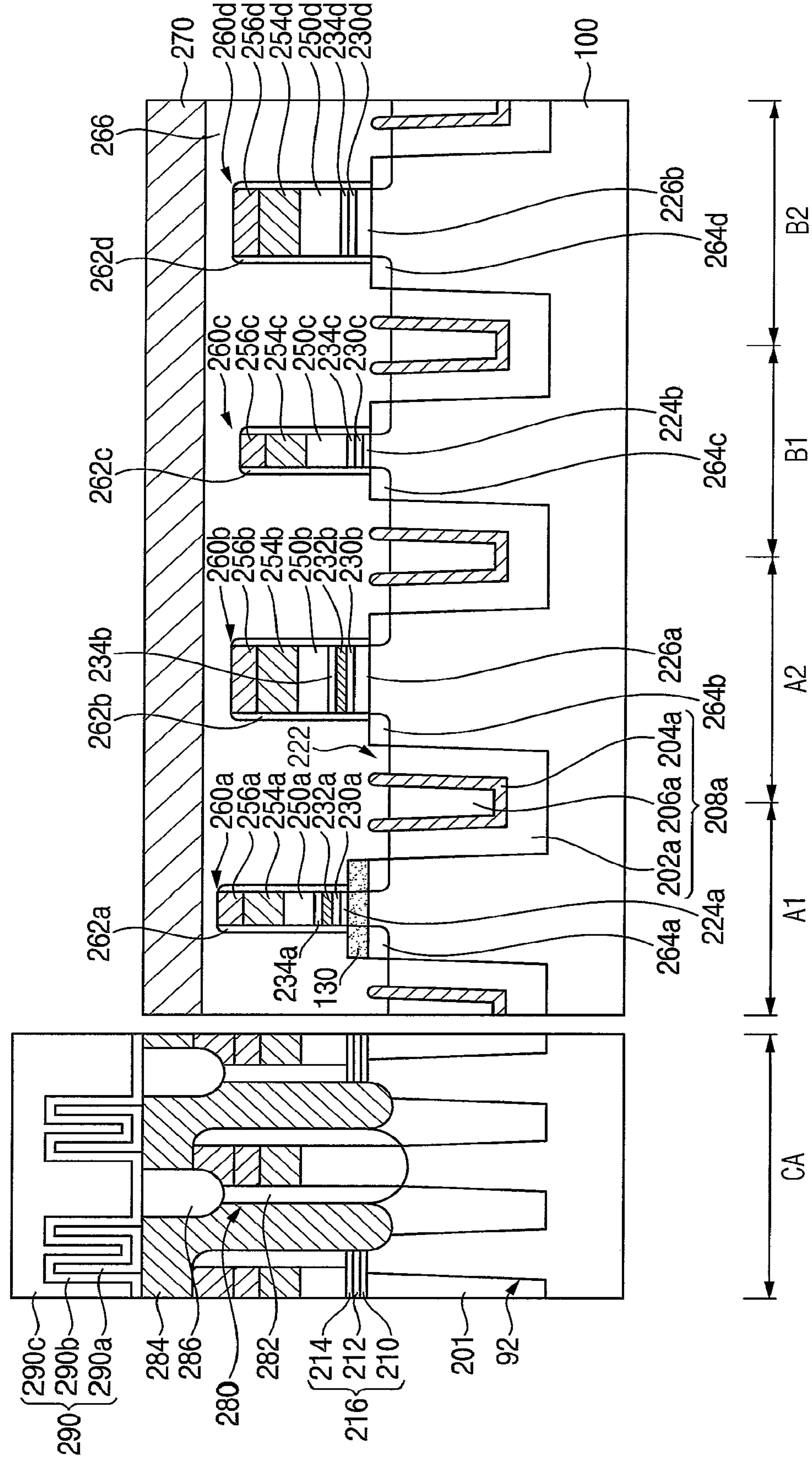
FIG. 17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224b
264c
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234b
226a
264b
A2
222
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206a
202a
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262a
224a
264a
234a
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216
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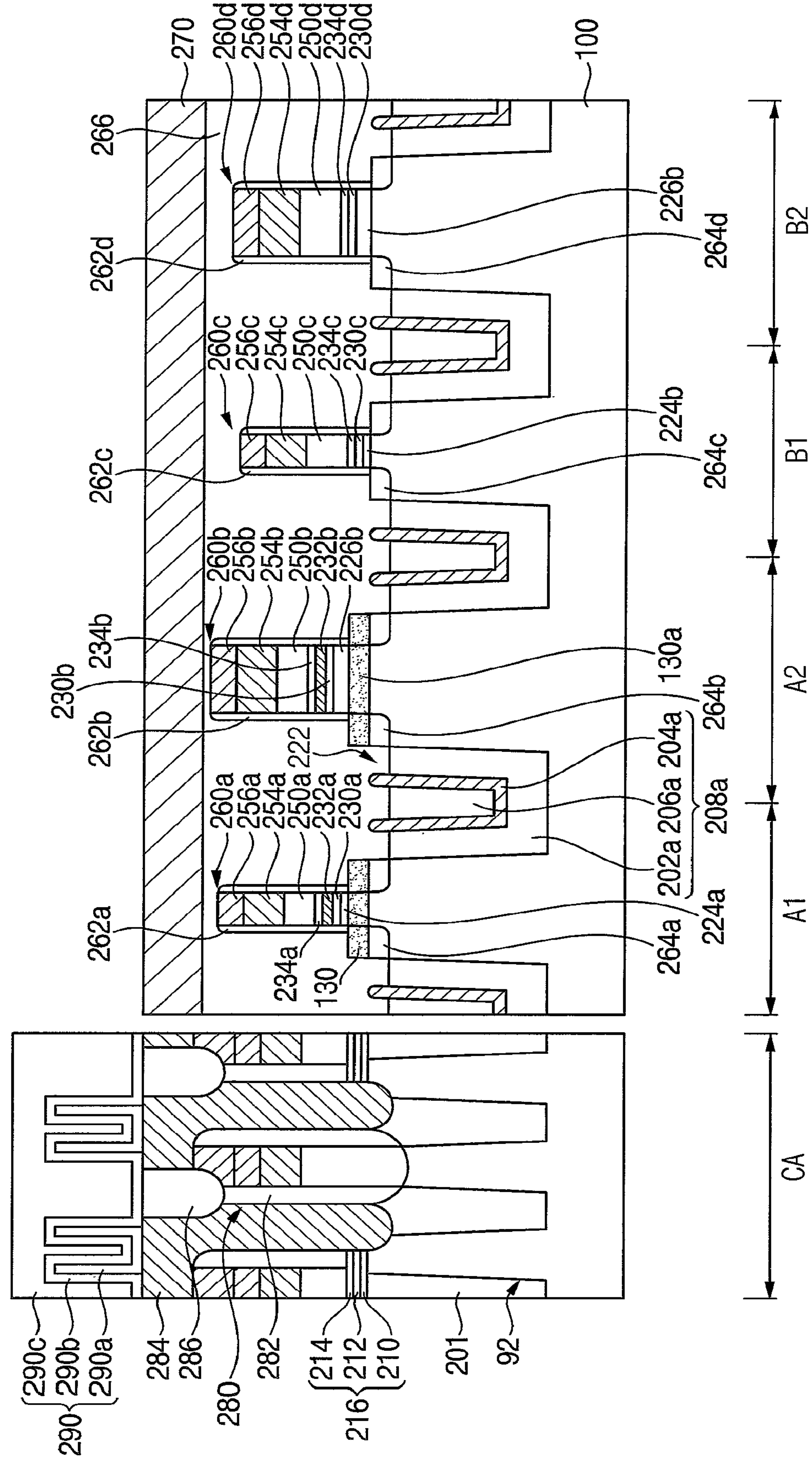
92

FIG. 18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226b
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264b
A2
262b
222
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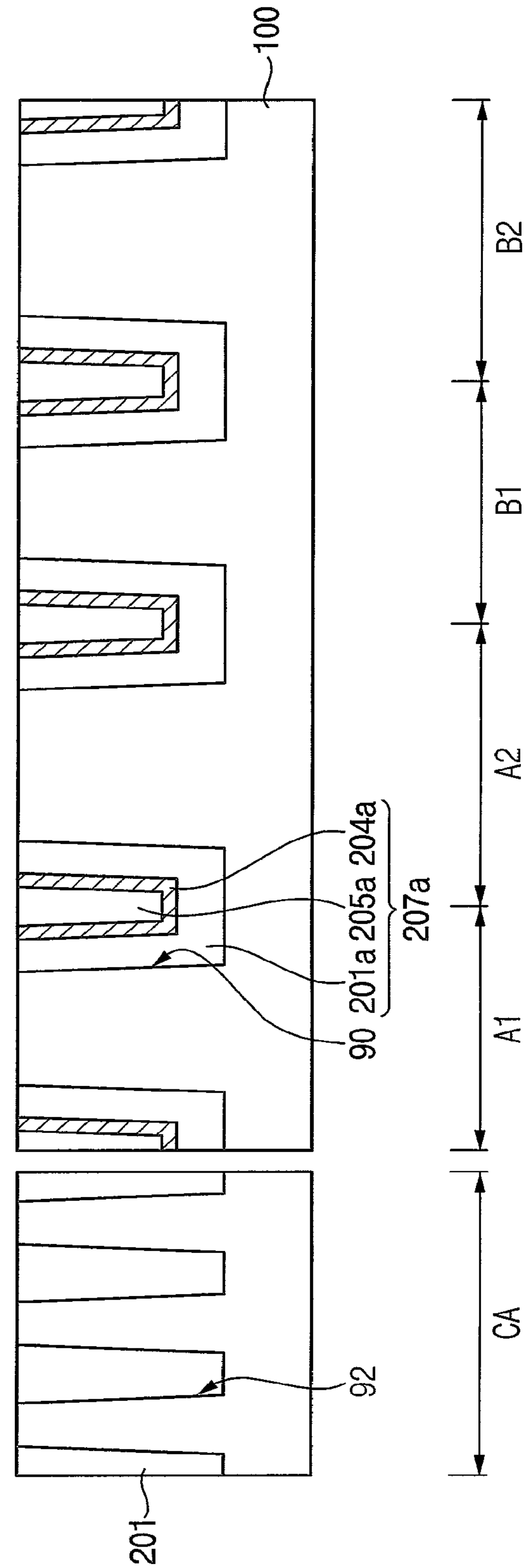
FIG. 19



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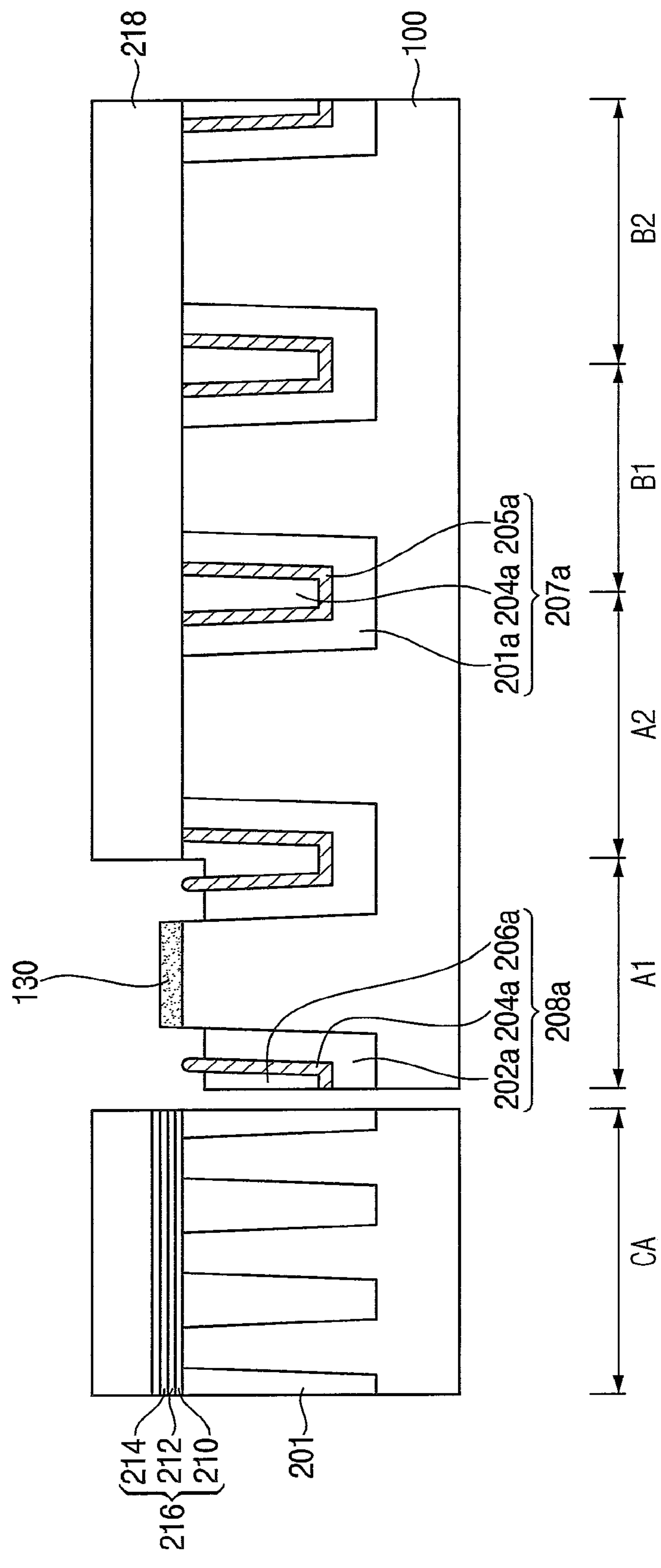
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FIG. 21

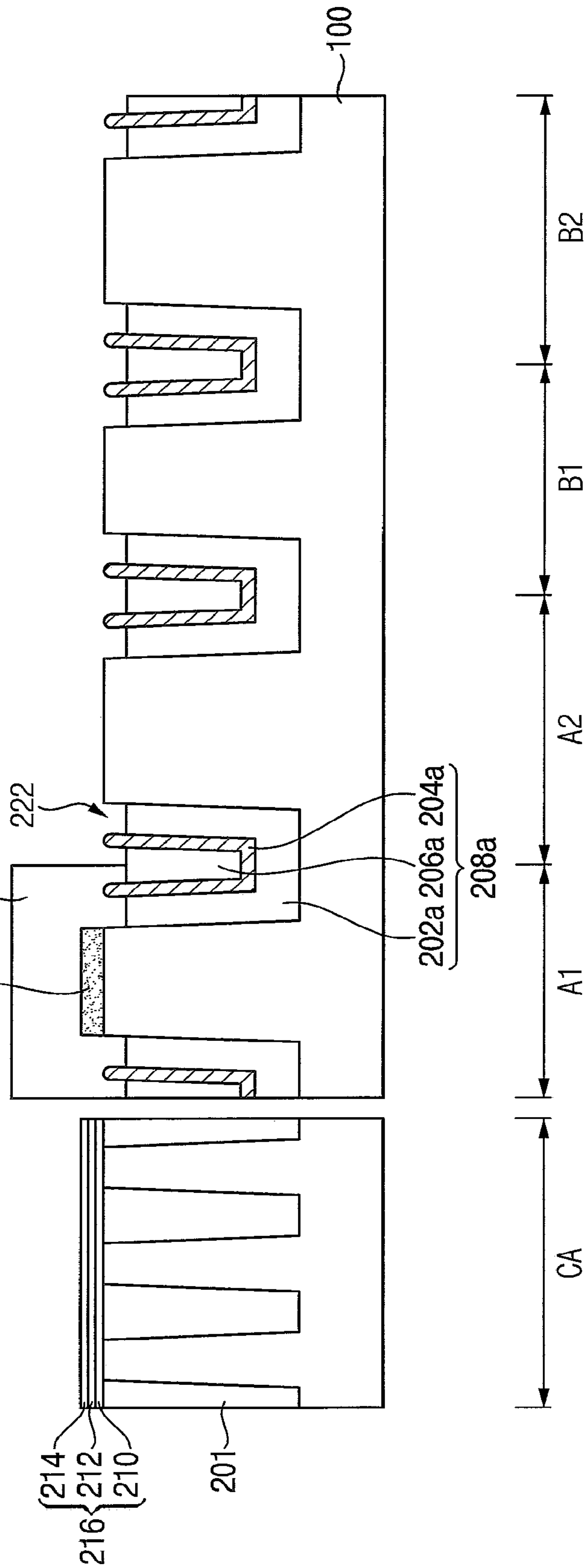
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FIG. 22

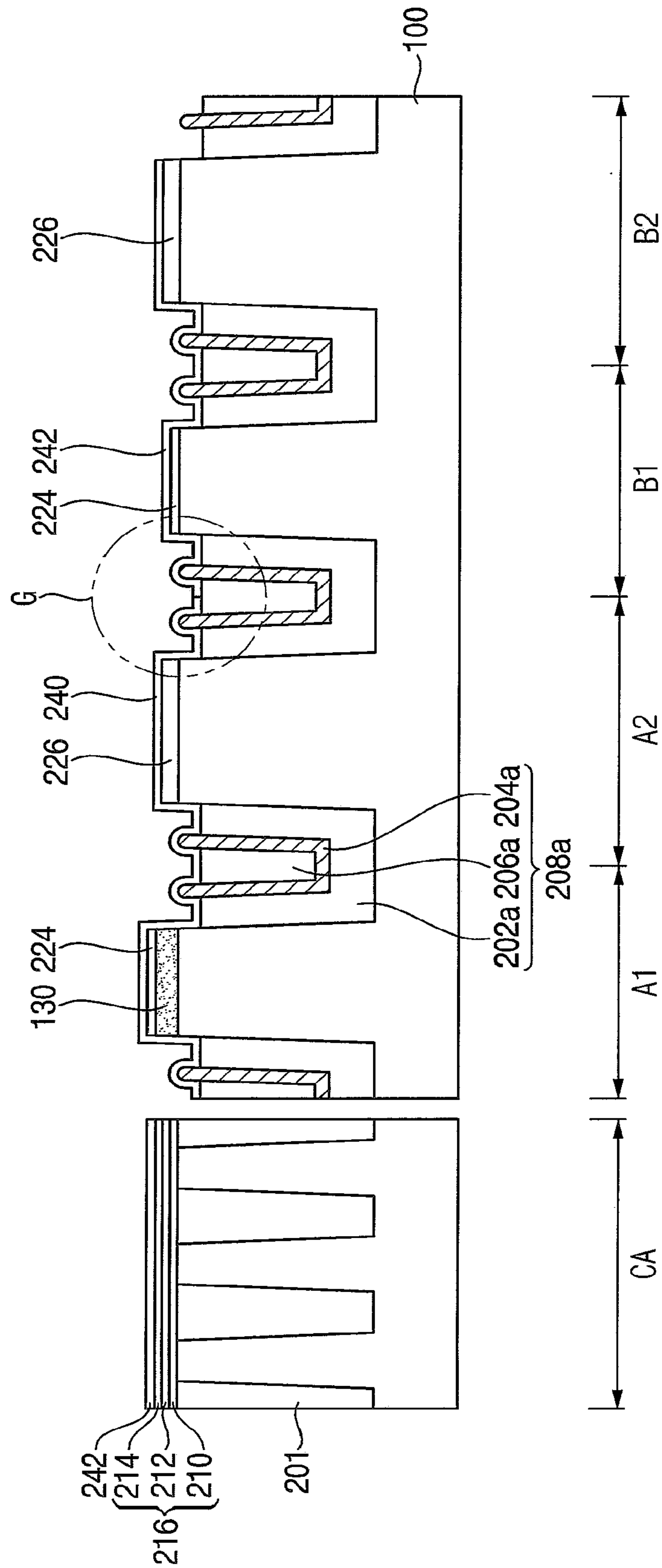
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FIG. 23

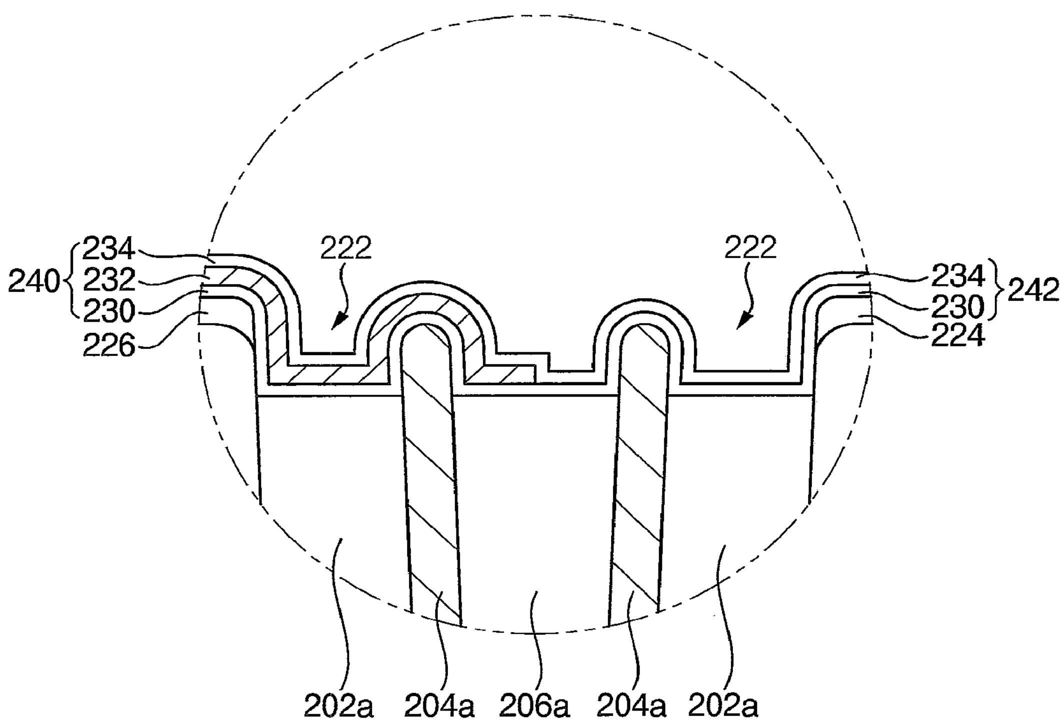
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FIG. 24

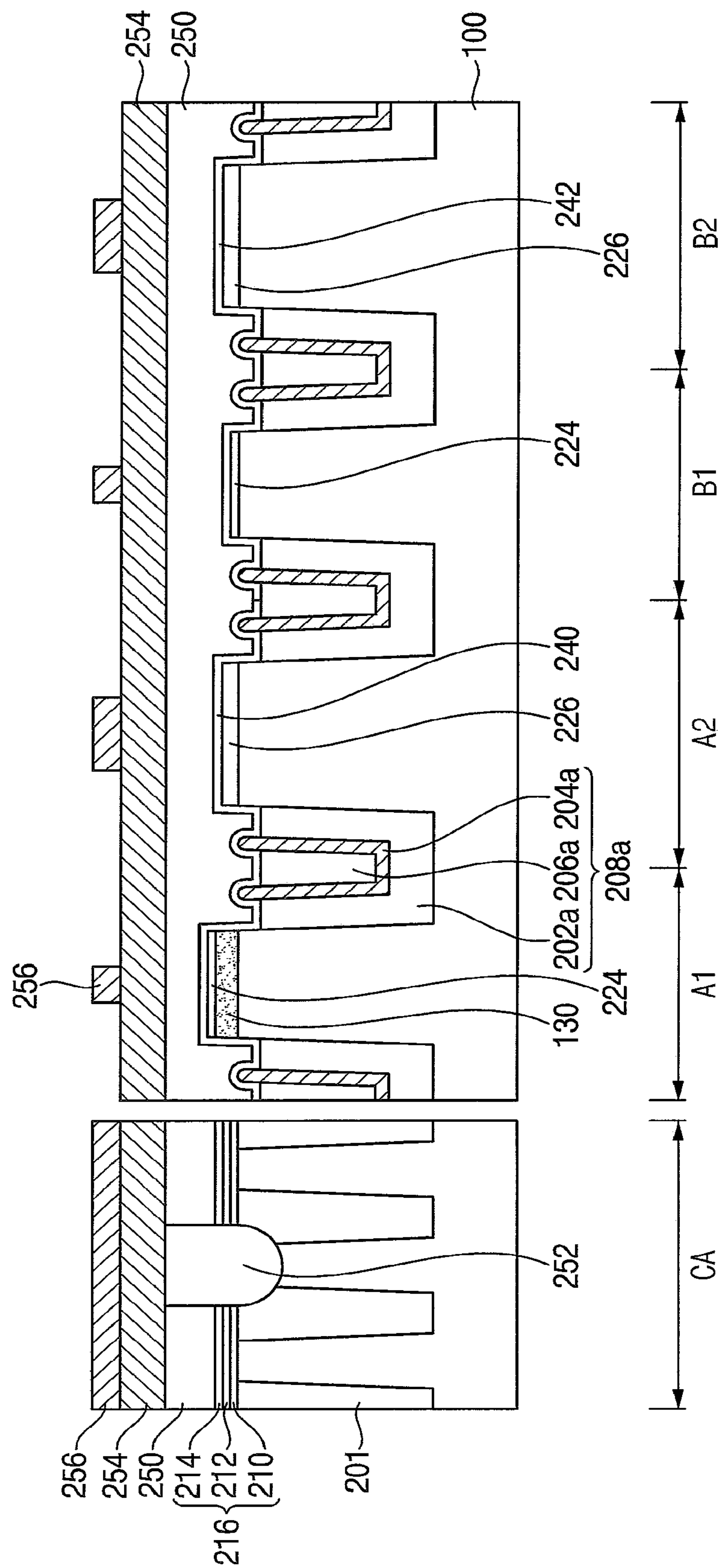
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FIG. 25

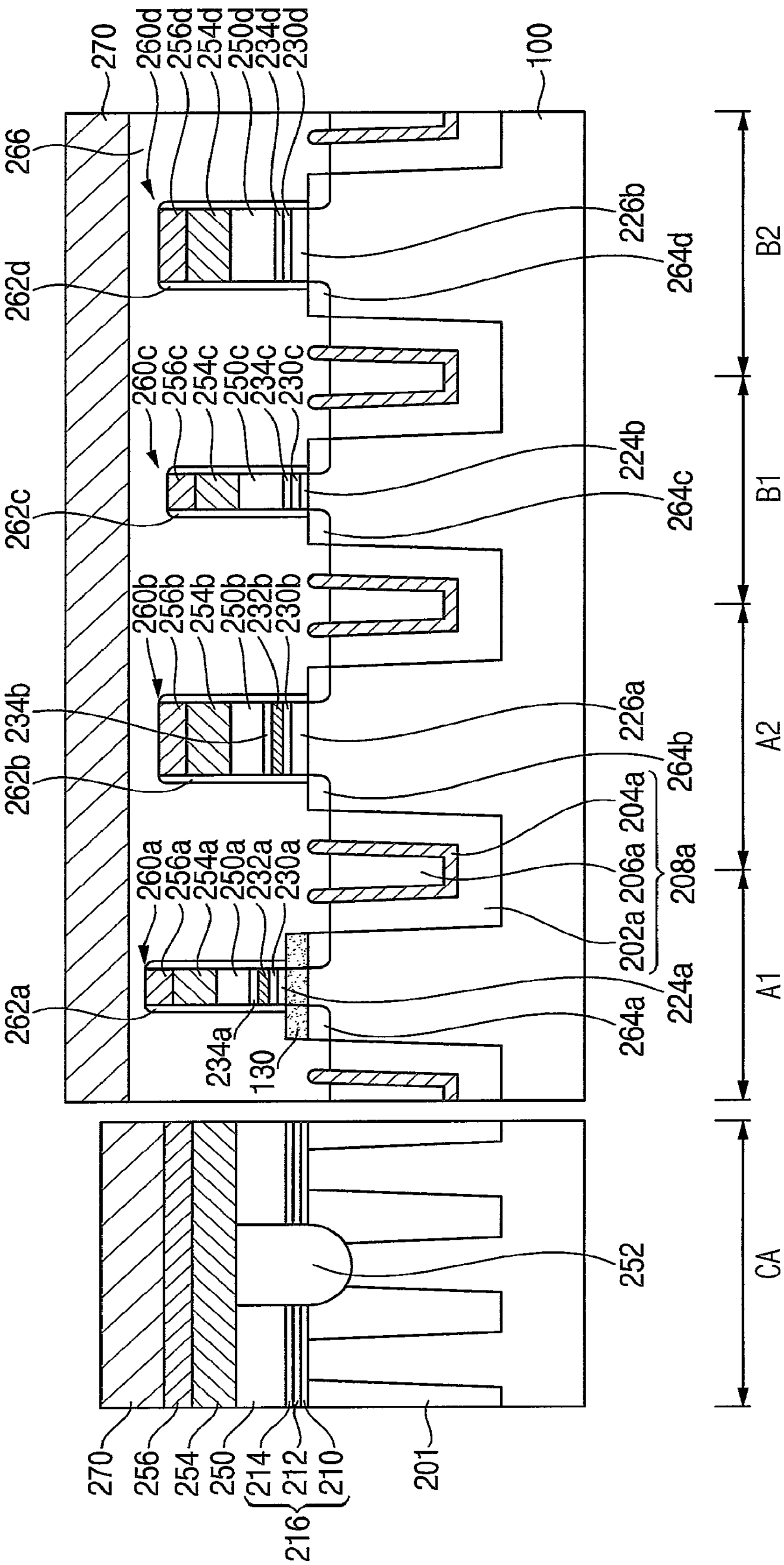
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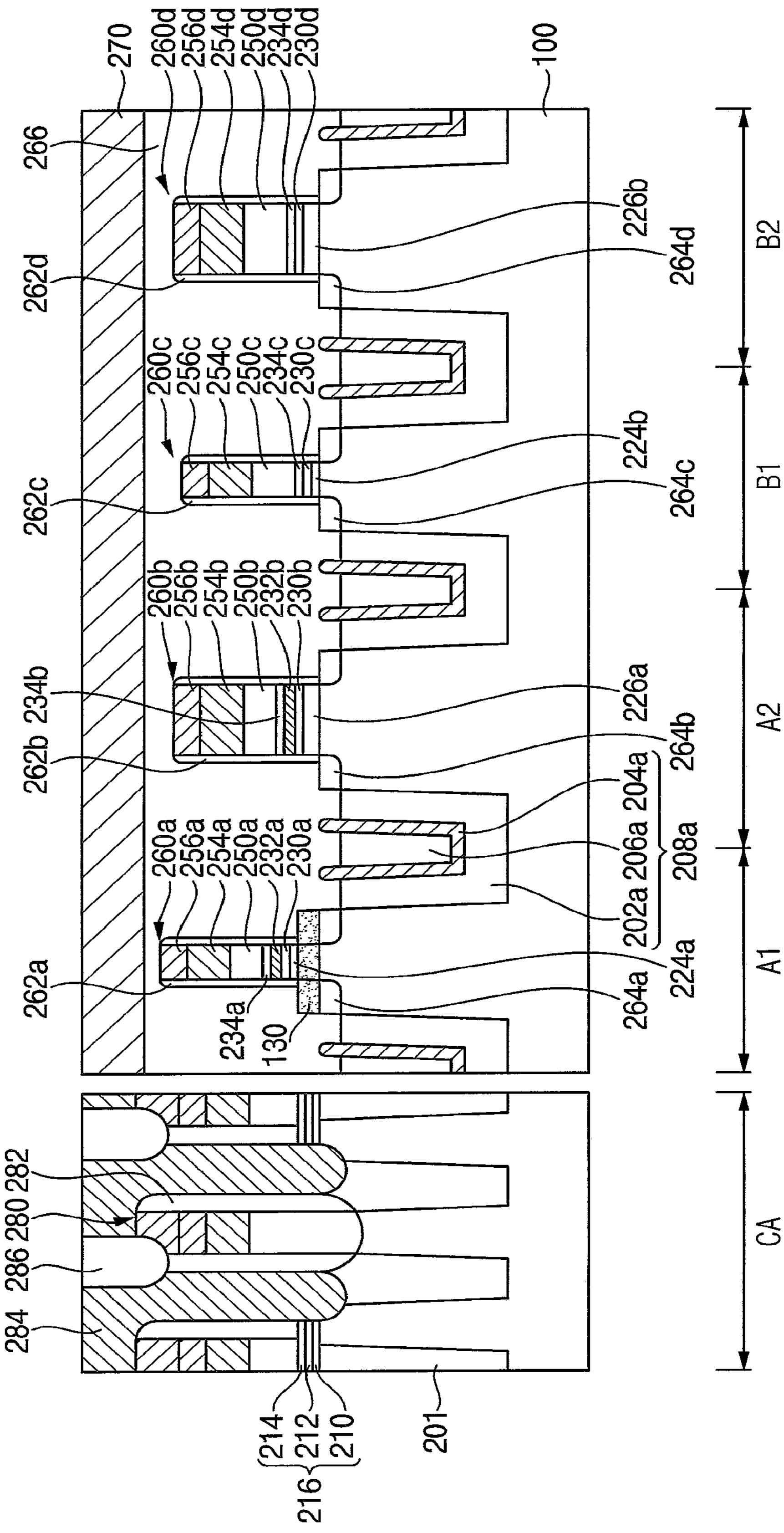
FIG. 26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SEMICONDUCTOR DEVICES

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. patent application Ser. No. 17/386,008 filed Jul. 27, 2021, which is incorporated by reference herein in its entirety.

Korean Patent Application No. 10-2020-0175579, filed on Dec. 15, 2020, in the Korean Intellectual Property Office, and entitled: "Semiconductor Devices," is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

Example embodiments relate to a semiconductor device. More particularly, example embodiments relate to a semiconductor device including transistors.

2. Description of the Related Art

A semiconductor device may include a plurality of transistors. The transistors included in the semiconductor device may have various structures according to required electrical performance, e.g., an operating voltage and/or driving currents. For example, an NMOS transistor and a PMOS transistor may have different stacked structures. For an isolation between NMOS transistors and an isolation between PMOS transistors, isolation structures may be formed at a substrate.

SUMMARY

According to example embodiments, there is provided a semiconductor device that may include a first trench formed at a substrate of a first region, a second trench formed at the substrate of a second region, a first isolation structure filling the first trench, a second isolation structure filling the second trench, a first gate structure formed on the substrate of the first region, and a second gate structure formed on the substrate of the second region. The first isolation structure may include a first inner wall oxide pattern, a first liner, and a first filling insulation pattern sequentially stacked. The second isolation structure may include a second inner wall oxide pattern, a second liner, and a second filling insulation pattern sequentially stacked. The first gate structure may include a first high-k dielectric pattern, a first P-type metal pattern, and a first N-type metal pattern sequentially stacked. The second gate structure may include a second high-k dielectric pattern and a second N-type metal pattern sequentially stacked. The first inner wall oxide pattern and the first liner may be conformally on a surface of the first trench. The first liner may protrude from upper surfaces of the first inner wall oxide pattern and the first filling insulation pattern. The second inner wall oxide pattern and the second liner may be conformally on a surface of the second trench. The second liner may protrude from upper surfaces of the second inner wall oxide pattern and the second filling insulation pattern.

According to example embodiments, there is provided a semiconductor device that may include a substrate including a cell array region, a first peripheral region, and a second peripheral region, memory cells on the substrate of the cell array region, trenches formed at the substrate of the first and second peripheral regions, an isolation structure filling each of the trenches, a channel layer on the substrate of the first

peripheral region, a first gate structure formed on the channel layer, and a second gate structure formed on the substrate of the second peripheral region. The isolation structure may include an inner wall oxide pattern, a nitride liner, and a filling insulation pattern sequentially stacked. The channel layer may include silicon germanium. The first gate structure may include a first high-k dielectric pattern, a first P-type metal pattern, and a first N-type metal pattern sequentially stacked. The second gate structure may include a second high-k dielectric pattern and a second N-type metal pattern sequentially stacked. The inner wall oxide pattern and the nitride liner may be conformally on surfaces of the trenches. The nitride liner may protrude from upper surfaces of the inner wall oxide pattern and the filling insulation pattern.

According to example embodiments, there is provided a semiconductor device that may include a substrate including a cell array region, a first peripheral region, a second peripheral region, a third peripheral region, and a fourth peripheral region, memory cells on the substrate of the cell array region, trenches at the substrate of the first to fourth peripheral regions, an isolation structure filling each of the trenches, a channel layer on the substrate of the first peripheral region, a first gate structure formed on the channel layer, a second gate structure formed on the substrate of the second peripheral region, a third gate structure formed on the substrate of the third peripheral region, and a fourth gate structure formed on the substrate of the fourth peripheral region. The isolation structure may include an inner wall oxide pattern, a nitride liner, and a filling insulation pattern sequentially stacked. The channel layer may include silicon germanium. The first gate structure may include a first interface insulation pattern, a first high-k dielectric pattern, a first P-type metal pattern, and a first N-type metal pattern sequentially stacked. The second gate structure may include a second interface insulation pattern, a second high-k dielectric pattern, a second P-type metal pattern, and a second N-type metal pattern sequentially stacked. A thickness of the second interface insulation pattern may be greater than a thickness of the first interface insulation pattern, The third gate structure may include a third interface insulation pattern, a third high-k dielectric pattern, and a third N-type metal pattern sequentially stacked. The fourth gate structure may include a fourth interface insulation pattern, a fourth high-k dielectric pattern, and a fourth N-type metal pattern sequentially stacked. The nitride liner may be conformally on the surfaces of the trenches. The nitride liner may protrude from upper surfaces of the inner wall oxide pattern and the filling insulation pattern. Heights of uppermost surfaces of the nitride liners are substantially the same.

BRIEF DESCRIPTION OF THE DRAWINGS

Features will become apparent to those of skill in the art by describing in detail exemplary embodiments with reference to the attached drawings, in which:

FIG. **1** is a cross-sectional view of a semiconductor device in accordance with example embodiments;

FIG. **2** is a cross-sectional view of a semiconductor device in accordance with example embodiments;

FIG. **3** is a cross-sectional view of a semiconductor device in accordance with example embodiments;

FIGS. **4** to **16** are cross-sectional views of stages in a method of manufacturing a semiconductor device in accordance with example embodiments;

FIG. **17** is a cross-sectional view of a semiconductor device in accordance with example embodiments;

FIG. **18** is a cross-sectional view of a semiconductor device in accordance with example embodiments; and

FIGS. **19** to **26** are cross-sectional views of stages in a method of manufacturing a semiconductor device in accordance with example embodiments.

DETAILED DESCRIPTION

FIG. **1** is a cross-sectional view of a semiconductor device in accordance with example embodiments.

Referring to FIG. **1**, a substrate **100** may include a first region A and a second region B. The substrate **100** may be, e.g., a single crystal silicon substrate or a silicon on insulator (SOI) substrate. The first region A may be a PMOS transistor region, and the second region B may be an NMOS transistor region.

A first trench **90***a* may be formed in the substrate **100** in the first region A, and a second trench **90***b* may be formed in the substrate **100** in the second region B. A first isolation structure **108***a* may be formed in the first trench **90***a*, and a second isolation structure **108***b* may be formed in the second trench **90***b*. In the substrate **100**, a region in which the first and second isolation structures **108***a* and **108***b* are disposed may serve as a field region. A region of the substrate **100** between the first and second isolation structures **108***a* and **108***b* may serve as an active region.

The first isolation structure **108***a* may include a first inner wall oxide pattern **102***a*, a first liner **104***a*, and a first filling insulation pattern **106***a*. The first inner wall oxide pattern **102***a* may be conformally formed on a surface of the first trench **90***a*, and the first liner **104***a* may be conformally formed on the first inner wall oxide pattern **102***a*. The first filling insulation pattern **106***a* may be formed on the first liner **104***a* to fill the first trench **90***a*.

The second isolation structure **108***b* may include a second inner wall oxide pattern **102***b*, a second liner **104***b*, and a second filling insulation pattern **106***b*. The second inner wall oxide pattern **102***b* may be conformally formed on a surface of the second trench **90***b*, and the second liner **104***b* may be conformally formed on the second inner wall oxide pattern **102***b*. The second filling insulation pattern **106***b* may be formed on the second liner **104***b* to fill the second trench **90***b*.

The first and second inner wall oxide patterns **102***a* and **102***b* may be formed by the same deposition process, and thus the first and second inner wall oxide patterns **102***a* and **102***b* may include the same material. For example, the first and second inner wall oxide patterns **102***a* and **102***b* may include silicon oxide. The first liner **104***a* and the second liner **104***b* may be formed by the same deposition process, and thus the first liner **104***a* and the second liner **104***b* may include the same material. For example, the first liner **104***a* and the second liner **104***b* may include silicon nitride. The first and second filling insulation patterns **106***a* and **106***b* may be formed by the same deposition process, and thus the first and second filling insulation patterns **106***a* and **106***b* may include the same material. For example, the first and second filling insulation patterns **106***a* and **106***b* may include silicon oxide.

In example embodiments, a thickness (i.e., a thickness in a vertical direction from the surface of the trench) of the first inner wall oxide pattern **102***a* may be greater than a thickness of the first liner **104***a*. A thickness of the second inner wall oxide pattern **102***b* may be greater than a thickness of the second liner **104***b*.

In example embodiments, each of the first and second inner wall oxide patterns **102***a* and **102***b* may have the thickness of about 200 angstroms or more. For example, each of the first and second inner wall oxide patterns **102***a* and **102***b* may have a thickness of about 250 angstroms to about 400 angstroms. If the thickness of each of the first and second inner wall oxide patterns **102***a* and **102***b* is less than 200 angstroms, an inner width of each of first and second recessed portions **110***a* and **110***b* subsequently described may be decreased.

In the first isolation structure **108***a*, the first liner **104***a* may protrude from, e.g., above, upper surfaces of the first inner wall oxide pattern **102***a* and the first filling insulation pattern **106***a*. That is, an uppermost surface of the first liner **104***a* may be higher than uppermost surfaces of the first inner wall oxide pattern **102***a* and the first filling insulation pattern **106***a*, e.g., relative to a bottom of the substrate **100**. A first recessed portion **110***a* may be formed, e.g., defined, between the first liner **104***a* and the substrate **100** adjacent thereto. For example, as illustrated in FIG. **1**, since the first liner **104***a* and a portion of the substrate **100** (e.g., an impurity region **170** in a top portion of the substrate **100**) protrude above the upper surface of the first inner wall oxide pattern **102***a* from opposite sides thereof, the first recessed portion **110***a* may be defined on the upper surface of the first inner wall oxide pattern **102***a* between the protruding portions of the first liner **104***a* and the substrate **100** (e.g., under dashed line in FIG. **1**).

In example embodiments, the uppermost surface of the first filling insulation pattern **106***a* may be substantially coplanar with the uppermost surface of the first inner wall oxide pattern **102***a*, or may be lower than the uppermost surface of the first inner wall oxide pattern **102***a*.

In example embodiments, the inner width of the first recessed portion **110***a*, e.g., a distance between directly facing surfaces of the first liner **104***a* and the substrate **100** along a horizontal direction parallel to the bottom of the substrate **100**, may be greater than a height from a bottom of the first recessed portion **110***a* to the uppermost surface of the first liner **104***a*. In example embodiments, the bottom of the first recessed portion **110***a* may be substantially flat, e.g., the upper surface of the first inner wall oxide pattern **102***a* defining the bottom of the first recessed portion **110***a* may be substantially flat. That is, a central portion of the bottom of the first recessed portion **110***a* may not have a rounded shape, and the bottom of the first recessed portion **110***a* may have a flat surface, e.g., an entire bottom surface of the first recessed portion **110** defined by the upper surface of the first inner wall oxide pattern **102***a* may be level and parallel to the bottom surface of the substrate **100**.

As such, an aspect ratio of the first recessed portion **110***a* may be decreased, and the inner width of the first recessed portion **110***a* may be increased. Therefore, metal layers deposited in the first recessed portion **110***a* may be easily removed in processing for manufacturing the semiconductor device. Thus, the metal layers may not remain in the first recessed portion **110***a*. Accordingly, defects (e.g., a bridge failure between gates) due to remains or residue of removed metal layers in the first recessed portion **110***a* may be decreased.

The second isolation structure **108***b* may have a shape and a structure similar to a shape and a structure of the first isolation structure **108***a*. That is, in the second isolation structure **108***b*, the second liner **104***b* may protrude from, e.g., above, upper surfaces of the second inner wall oxide pattern **102***b* and the second filling insulation pattern **106***b*. The uppermost surface of the second liner **104***b* may be higher than the uppermost surfaces of the second inner wall oxide pattern **102***b* and the second filling insulation pattern

106*b*, so a second recessed portion **110***b* may be formed between the second liner **104***b* and the substrate **100** adjacent thereto. In example embodiments, the uppermost surface of the second filling insulation pattern **106***b* may be substantially coplanar with the uppermost surface of the second inner wall oxide pattern **102***b*, or may be lower than the uppermost surface of the second inner wall oxide pattern **102***b*.

In example embodiments, the inner width of the second recessed portion **110***b* may be greater than a height from a bottom of the second recessed portion **110***b* to the uppermost surface of the second liner **104***b*. In example embodiments, the bottom of the second recessed portion **110***b* may be substantially flat. That is, a central portion of the bottom of the second recessed portion **110***b* may not have a rounded shape, and the bottom of the second recessed portion **110***b* may have a flat surface.

In example embodiments, in the first and second isolation structures **108***a* and **108***b*, heights (e.g., vertical levels relative to the bottom of the substrate **100**) of the uppermost surfaces of the first and second liners **104***a* and **104***b* including silicon nitride may be substantially the same. For example, in the first and second isolation structures **108***a* and **108***b*, the uppermost surfaces of the first and second liners **104***a* and **104***b* including silicon nitride may be substantially coplanar with an upper surface of a portion of the substrate adjacent to the first and second liners **104***a* and **104***b*.

In example embodiments, the uppermost surface of the first inner wall oxide pattern **102***a* may be lower than the upper surface of the substrate **100** adjacent to the first trench **90***a*, and the uppermost surface of the second inner wall oxide pattern **102***b* may be lower than the upper surface of the substrate **100** adjacent to the second trench **90***b*. The uppermost surface of the first inner wall oxide pattern **102***a* and the uppermost surface of the second inner wall oxide pattern **102***b* may be substantially coplanar with each other.

In example embodiments, a vertical level of the bottom of the second recessed portion **110***b* may be substantially the same as a vertical level of the bottom of the first recessed portion **110***a*. Further, a height of a protruding portion of the first liner **104***a* may be substantially the same as a height of a protruding portion of the second liner **104***b*.

A channel layer **130** may be formed on the substrate **100** of the first region A. A lattice constant of the channel layer **130** may be greater than a lattice constant of the substrate **100**. For example, the channel layer **130** may include silicon germanium.

Hole mobility of a PMOS transistor formed on the channel layer **130** may be increased by the channel layer **130**. Further, a threshold voltage of the PMOS transistor may be controlled by controlling a work function of the channel layer **130**. For example, the channel layer **130** may have a low work function, so that the PMOS transistor may have a target threshold voltage. A first gate structure **160** may be formed on the channel layer **130**.

First impurity regions **170** serving as first source/drain regions may be formed at the channel layer **130** and the substrate **100** adjacent to both sides of the first gate structure **160**. The first impurity regions **170** may be doped with impurities, e.g., P-type impurities.

A second gate structure **162** may be formed on the substrate **100** of the second region B. Second impurity regions **172** serving as second source/drains may be formed in the substrate **100** adjacent to both sides of the second gate structure **162**. The second impurity regions **172** may be doped with impurities, e.g., N-type impurities.

The first gate structure **160** may include a first interface insulation pattern, a first high-k dielectric pattern **140***a*, a first P-type metal pattern **143**, a first N-type metal pattern **144***a*, a first lower electrode **146***a*, a first barrier pattern **148***a*, a first upper electrode **150***a*, and a first capping layer pattern **152***a* sequentially stacked. The second gate structure **162** may include a second interface insulation pattern, a second high-k dielectric pattern **140***b*, a second N-type metal pattern **144***b*, a second lower electrode **146***b*, a second barrier pattern **148***b*, a second upper electrode **150***b*, and a second capping layer pattern **152***b* sequentially stacked.

That is, the first gate structure **160** may include the first P-type metal pattern **143** and the first N-type metal pattern **144***a* sequentially stacked, while the second gate structure **162** may include the second N-type metal pattern **144***b* (i.e., without a p-type metal pattern). Except for stacking of the P-type metal pattern and the N-type metal pattern, stacked structures of the first and second gate structures **160** and **162** may be the same.

As the P-type metal pattern is not included in the second gate structure **162**, a thickness of the first gate structure **160** may be greater than a thickness of the second gate structure **162**, e.g., in the vertical direction along a normal to the bottom of the substrate **100**. That is, the first gate structure **160** may have a first thickness, and the second gate structure **162** may have a second thickness less than the first thickness.

The first and second interface insulation patterns may include, e.g., silicon oxide or silicon oxynitride. The first high-k dielectric pattern **140***a* and the second high-k dielectric pattern **140***b* may include a metal oxide having a dielectric constant higher than a dielectric constant of silicon oxide. For example, The first high-k dielectric pattern **140***a* and the second high-k dielectric pattern **140***b* may include at least one of hafnium oxide (HfO), hafnium silicate (HfSiO), hafnium oxide nitride (HfON), hafnium silicon oxynitride (HfSiON), lanthanum oxide (LaO), lanthanum aluminum oxide, (LaAlO), zirconium oxide (ZrO), Zirconium silicate (ZrSiO), zirconium oxide nitride (ZrON), zirconium silicon oxynitride (ZrSiON), tantalum oxide (TaO), titanium oxide (TiO), barium strontium titanium oxide (BaSrTiO), barium titanium oxide (BaTiO), strontium titanium oxide (SrTiO), yttrium oxide (YO), aluminum oxide (Al2O3), and lead scandium tantalum oxide (PbScTaO). For example, the first and second high-k dielectric patterns **140***a* and **140***b* may include HfO2, Al2O3, HfAlO3, Ta2O3, or TiO2. In an embodiment, the first high-k dielectric pattern of the first gate structure includes an oxide including hafnium (Hf) and silicon (Si), and the second high-k dielectric pattern of the second gate structure includes an oxide including hafnium (Hf), silicon (Si), and aluminum (Al).

The first P-type metal pattern **143** may include a metal material having P-work function. In example embodiments, the first P-type metal pattern **143** may include at least one of aluminum, aluminum oxide, titanium nitride, tungsten nitride, and ruthenium oxide. For example, the first P-type metal pattern **143** may include a stacked structure including a titanium nitride layer, an aluminum layer, and a titanium nitride layer or a stacked structure including a titanium nitride layer, an aluminum oxide layer, and a titanium nitride layer.

The first N-type metal pattern **144***a* and the second N-type metal pattern **144***b* may include a metal having an N-work function. In example embodiments, the N-type metal pattern **25** may include at least one of lanthanum (La), lanthanum oxide (LaO), tantalum (Ta), tantalum nitride (TaN), niobium (Nb), and titanium nitride (TiN). For example, each of the first and second N-type metal patterns **144**_a_ and **144**_b_ may include a stacked structure including a lanthanum layer and a titanium nitride layer or a stacked structure including a lanthanum oxide layer and a titanium nitride layer.

The first lower electrode **146**_a_ and the second lower electrode **146**_b_ may include polysilicon doped with impurities. The first lower electrode **146**_a_ may be doped with P-type impurities, and the second lower electrode **146**_b_ may be doped with N-type impurities.

The first and second barrier patterns **148**_a_ and **148**_b_ may include, e.g., titanium nitride, tungsten nitride, or tantalum nitride.

The first upper electrode **150**_a_ and the second upper electrode **150**_b_ may include a metal. The first upper electrode **150**_a_ may include, e.g., a metal silicide layer, a metal nitride layer, or a metal layer. The metal silicide may include, e.g., cobalt silicide or tungsten silicide. The metal nitride layer may include, e.g., a titanium nitride layer or a tantalum nitride layer. The metal layer may include, e.g., tungsten.

The first and second capping layer patterns **152**_a_ and **152**_b_ may include, e.g., silicon nitride. A first spacer **164** may be formed on a sidewall of the first gate structure **160**. A second spacer **166** may be formed on a sidewall of the second gate structure **162**. The first and second spacers **164** and **166** may include, e.g., silicon oxide and/or silicon nitride.

In the semiconductor device, the first and second isolation structures **108**_a_ and **108**_b_ may include first and second liners **104**_a_ and **104**_b_ having protruding portions, respectively, that define first and second recessed portions **110**_a_ and **110**_b_ adjacent to the first and second gate structures **160** and **162**, respectively. However, during manufacturing, metal material may not remain in the first and second recessed portions **110**_a_ and **110**_b_ due to increased width thereof. Therefore, in the semiconductor device, potential defects caused by residual metal materials in the first and second recessed portions **110**_a_ and **110**_b_ may be decreased.

FIG. **2** is a cross-sectional view illustrating a semiconductor device in accordance with example embodiments. The semiconductor device in FIG. **2** is substantially the same as the semiconductor device described with reference to FIG. **1**, except for shapes of the first and second isolation structures.

Referring to FIG. **2**, a bottom of each of the first and second recessed portions **110**_a_ and **110**_b_ may have a slope with respect to a flat upper surface of the substrate **100**, e.g., the bottom of each of the first and second recessed portions **110**_a_ and **110**_b_ may be inclined at an oblique angle with respect to the bottom of the substrate **100**. A central portion of the bottom of each of the first and second recessed portions **110**_a_ and **110**_b_ may not have a rounded shape.

In example embodiments, an uppermost surface of the first filling insulation pattern **106**_a_ may be substantially coplanar with an upper surface of the first inner wall oxide pattern **102**_a_, or may be lower than the upper surface of the first inner wall oxide pattern **102**_a_. In example embodiments, an uppermost surface of the second filling insulation pattern **106**_b_ may be substantially coplanar with an upper surface of the second inner wall oxide pattern **102**_b_, or may be lower than the upper surface of the second inner wall oxide pattern **102**_b_.

FIG. **3** is a cross-sectional view illustrating a semiconductor device in accordance with example embodiments. The semiconductor device in FIG. **3** is substantially the same as the semiconductor device described with reference to FIG. **1**, except for shapes of the first and second isolation structures.

Referring to FIG. **3**, a width of the first filling insulation pattern **106**_a_, e.g., as measured between directly facing surfaces of the first liner **104**_a_, may be greater than a width of the first inner wall oxide pattern **102**_a_. In example embodiments, the width of the first filling insulation pattern **106**_a_ may be greater than twice the width of the first inner wall oxide pattern **102**_a_. The width of the first inner wall oxide pattern **102**_a_ may be a thickness of the first inner wall oxide pattern **102**_a_ in the vertical direction from a sidewall of the first trench **90**_a_, e.g., the width of the first inner wall oxide pattern **102**_a_ may be a distance between directly facing surfaces of the first inner wall oxide pattern **102**_a_ and the sidewall of the first trench **90**_a_ along a horizontal direction parallel to the bottom of the substrate **100**.

When the width of the first filling insulation pattern **106**_a_ is greater than the width of the first inner wall oxide pattern **102**_a_, the first filling insulation pattern **106**_a_ may be easily exposed to an etching gas during an etching process. Thus, an amount of etching of the first filling insulation pattern **106**_a_ may increase, so that an uppermost surface of the first filling insulation pattern **106**_a_ may be lower than an uppermost surface of the first inner wall oxide pattern **102**_a_, e.g., relative to the bottom of the substrate **100**.

A width of the second filling insulation pattern **106**_b_ may be greater than a width of the second inner wall oxide pattern **102**_b_. In this case, an uppermost surface of the second filling insulation pattern **106**_b_ may be lower than an uppermost surface of the second inner wall oxide pattern **102**_b_. In example embodiments, the width of the second filling insulation pattern **106**_b_ may be greater than twice the width of the second inner wall oxide pattern **102**_b_.

FIGS. **4** to **16** are cross-sectional views of stages in a method of manufacturing a semiconductor device in accordance with example embodiments.

Referring to FIG. **4**, the substrate **100** may include the first region A and the second region B. A portion of the substrate **100** may be etched to form the first and second trenches **90**_a_ and **90**_b_. The first trench **90**_a_ may be formed in a portion of the substrate **100** of the first region A, and the second trench **90**_b_ may be formed in a portion of the substrate **100** of the second region B.

An inner wall oxide layer may be conformally formed on an upper surface of the substrate **100** and surfaces of the first and second trenches **90**_a_ and **90**_b_. A liner layer may be formed on the inner wall oxide layer. The inner wall oxide layer may be formed to have a thickness greater than a thickness of the liner layer. The inner wall oxide layer may have a thickness of 200 angstroms or more, e.g., the inner wall oxide layer may have a thickness of about 250 angstroms to about 400 angstroms.

In example embodiments, the inner wall oxide layer may include silicon oxide. The inner wall oxide layer may be formed by a thermal oxidation process and/or a deposition process. In example embodiments, the liner layer may include silicon nitride. The liner layer may be formed by a deposition process.

A filling insulation layer may be formed on the liner layer to fill the first and second trenches **90**_a_ and **90**_b_. The filling insulation layer may include silicon oxide. The filling insulation layer may be formed by a deposition process.

Thereafter, the filling insulation layer, the liner layer, and the inner wall oxide layer may be planarized until an upper surface of the substrate **100** is exposed to form a preliminary first isolation structure **107**_a_ and a preliminary second isolation structure **107**_b_. The preliminary first isolation structure **107**a may be formed in the first trench **90**a, and the preliminary second isolation structure **107**b may be formed in the second trench **90**b.

The preliminary first isolation structure **107**a may include a preliminary first inner wall oxide pattern **101**a, the first liner **104**a, and a preliminary first filling insulation layer pattern **105**a. The preliminary second isolation structure **107**b may include a preliminary second inner wall oxide pattern **101**b, the second liner **104**b, and a preliminary second filling insulation layer pattern **105**b.

Referring to FIG. **5**, a first mask layer **120** may be formed to cover the substrate **100** and the preliminary first and second isolation structures **107**a and **107**b. The first mask layer **120** may be a silicon oxide layer. For example, the first mask layer **120** may include TEOS (Tetra Ethyl Ortho Silicate) material.

A first photoresist pattern **122** may be formed on the first mask layer **120**. The first photoresist pattern **122** may cover the second region B, and may expose the first region A.

Referring to FIG. **6**, the first mask layer may be etched using the first photoresist pattern **122** as an etching mask to form a first mask pattern **120**a. The etching process of the first mask layer may include a wet etching process. When the wet etching process is performed, etching damages applied to the surface of the substrate **100** may be decreased. Thereafter, the first photoresist pattern **122** may be removed.

As the first mask layer includes silicon oxide, a silicon oxide layer may be etched in the etching process for forming the first mask pattern **120**a. In the etching process for forming the first mask pattern **120**a, a layer positioned below an etched first mask pattern **120**a may be overetched, and thus a silicon oxide layer of an upper portion of the preliminary first isolation structure **107**a may be etched together.

Therefore, upper portions of the preliminary first inner wall oxide pattern **101**a and the preliminary first filling insulation layer pattern **105**a included in the preliminary first isolation structure **107**a may be partially etched to form the first isolation structure **108**a. The first isolation structure **108**a may include the first inner wall oxide pattern **102**a, the first liner **104**a, and the first filling insulation pattern **106**a.

In the first isolation structure **108**a, an upper surface of the first inner wall oxide pattern **102**a may be lower than an upper surface of the first liner **104**a, so that the first recessed portion **110**a may be formed between the first liner **104**a and the substrate **100** adjacent thereto. The substrate **100** of the first region A and an upper surface of the first isolation structure **108**a may be exposed by the first mask pattern **120**a.

In example embodiments, an etching thickness in the vertical direction of the preliminary first inner wall oxide pattern **101**a may be less than a depositing thickness (i.e., a depositing thickness in the vertical direction from a sidewall of the first trench **90**a) of the inner wall oxide layer. In example embodiments, an inner width of the first recessed portion **110**a may be greater than a height from a bottom of the first recessed portion **110**a to an uppermost surface of the first liner **104**a.

As the first liner **104**a includes silicon nitride, the first liner **104**a may be hardly removed in the etching process. Thus, the uppermost surface of the first liner **104**a may be substantially coplanar with the upper surface of the substrate **100** adjacent to the first liner **104**a.

In example embodiments, an upper portion, e.g., an uppermost surface, of the first liner **104**a may have a rounded shape, after the etching process. Further, an edge of the active region of the substrate **100** may have a rounded shape, after the etching process.

The first liner **104**a may protrude from, e.g., above, upper portions of the first inner wall oxide pattern **102**a and the first filling insulation pattern **106**a. That is, the uppermost surface of the first liner **104**a may be higher than the uppermost surfaces of the first inner wall oxide pattern **102**a and the first filling insulation pattern **106**a.

In example embodiments, a bottom of the first recessed portion **110**a may have a flat surface. As the inner width of the first recessed portion **110**a is increased, a central portion of the bottom of the first recessed portion **110**a may not have a rounded shape. The central portion of the bottom of the first recessed portion **110**a may have a flat surface. In some example embodiments, as shown in FIG. **2**, the bottom of the first recessed portion **110**a may have a slope with respect to the upper flat surface of the substrate.

Referring to FIG. **7**, the channel layer **130** may be formed on the upper surface of the substrate **100** of the first region exposed by the first mask pattern **120**a. The channel layer **130** may be formed by, e.g., a selective epitaxial growth process (SEG). The channel layer **130** may include silicon germanium. The channel layer **130** may be selectively formed on the substrate **100** for forming a PMOS transistor.

Referring to FIG. **8**, a second photoresist pattern **124** may be formed on the channel layer **130** and the first isolation structure **108**a in the first region A. The second photoresist pattern **124** may cover the first region A, and may expose the second region B.

The first mask pattern **120**a in the second region B may be selectively exposed by the second photoresist pattern **124**. The channel layer **130** and the first isolation structure **108**a in the first region A may not be exposed by the second photoresist pattern **124**.

Referring to FIG. **9**, the first mask pattern **120**a may be removed using the second photoresist pattern **124** as an etching mask. The removing process of the first mask pattern **120**a may include a wet etching process.

As the first mask pattern **120**a includes silicon oxide, a silicon oxide layer may be etched in the removing process of the first mask pattern **120**a. In the removing process of the first mask pattern **120**a, a silicon oxide layer positioned below a removed first mask pattern **120**a may be overetched, and thus a silicon oxide layer of an upper portion of the preliminary second isolation structure **107**b may be etched together.

Therefore, upper portions of the preliminary second inner wall oxide pattern **101**b and the preliminary second filling insulation layer pattern **105**b included in the preliminary second isolation structure **107**b may be partially etched to form the second isolation structure **108**b. However, as the second liner **104**b includes silicon nitride, the second liner **104**b may be hardly removed during the removing process.

The second isolation structure **108**b may include the second inner wall oxide pattern **102**b, the second liner **104**b, and the second filling insulation pattern **106**b. The second liner **104**b may protrude from, e.g., above, upper portions of the second inner wall oxide pattern **102**b and the second filling insulation pattern **106**b. That is, the uppermost surface of the second liner **104**b may be higher than uppermost surfaces of the second inner wall oxide pattern **102**b and the second filling insulation pattern **106**b. The second recessed portion **110**b may be formed between the second liner **104**b and the substrate **100** adjacent thereto.

In example embodiments, in the removing process, a vertical level of a bottom of the second recessed portion **110**b may be substantially the same as a vertical level of the bottom of the first recessed portion **110**a. In example embodiments, a height of a protruding portion of the first liner **104**a may be substantially the same as a height of a protruding portion of the second liner **104**b.

In example embodiments, the second liner **104**b may be hardly etched during the etching processes, so that the uppermost surface of the second liner **104**b may be substantially coplanar with an upper surface of the substrate adjacent to the second liner **104**b. Upper surfaces of the first and second liners **104**a and **104**b may be substantially coplanar with each other.

In example embodiments, an inner width of the second recessed portion **110**b may be greater than a height from a bottom of the second recessed portion **110**b to an uppermost surface of the second liner **104**b. In example embodiments, the bottom of the second recessed portion **110**b may be substantially flat. That is, the central portion of the bottom of the second recessed portion **110**b may not have a rounded shape, and the bottom of the second recessed portion **110**b may have a flat surface. In some embodiments, as shown in FIG. **2**, the bottom of the second recessed portion **110**b may have a slope with respect to the flat upper surface of the substrate **100**.

As the second photoresist pattern **124** may be formed, a silicon oxide layer positioned at the first region A may not be removed during the removing process of the first mask pattern **120**a. In the removing process of the first mask pattern **120**a, layers included in the first isolation structure **108**a may not be removed, so that a shape of the first isolation structure **108**a may not change. Thereafter, the second photoresist pattern **124** may be removed.

Referring to FIG. **10**, a first interface insulation layer may be formed on the channel layer **130** of the first region A, and a second interface insulation layer may be formed on the substrate **100** of the second region B. The first and second interface insulation layers may be formed by a thermal oxidation process and/or a deposition process. The first and second interface insulation layers may be formed of, e.g., a silicon oxide layer and/or a silicon oxynitride layer.

A high-k dielectric layer **140** may be conformally formed on the first interface insulation layer, the second interface insulation layer, and the first and second isolation structures **108**a and **108**b. A P-type metal layer **142** may be conformally formed on the high-k dielectric layer **140**. The P-type metal layer **142** may be formed not to completely fill the first and second recessed portions **110**a and **110**b, and the P-type metal layer **142** may be formed along sidewalls and bottoms of the first and second recessed portions **110**a and **110**b. Thus, after forming the P-type metal layer **142**, each of the first and second recessed portions **110**a and **110**b may have remaining inner space.

The P-type metal layer **142** may include a metal layer having a P-work function. In example embodiments, the P-type metal layer **142** may include at least one of aluminum (Al), aluminum oxide, titanium nitride (TiN), tungsten nitride (WN), and ruthenium oxide (RuO_2). For example, the P-type metal layer **142** may include a stacked structure including a titanium nitride layer, an aluminum layer, and a titanium nitride layer or a stacked structure including a titanium nitride layer, an aluminum oxide layer, and a titanium nitride layer.

Referring to FIG. **11**, a third mask pattern **132** may be formed on the P-type metal layer **142**. The third mask pattern **132** may cover the first region A, and may expose the second region B. The third mask pattern **132** may include at least one of a photoresist, an amorphous carbon layer (ACL), a spin on hardmask (SOH), a spin on carbon (SOC), and a silicon nitride layer.

Referring to FIG. **12**, the P-type metal layer **142** of the second region B may be etched using the third mask pattern **132** as an etching mask so as to expose the high-k dielectric layer **140**. The P-type metal layer **142** may only remain in the first region A, and thus a preliminary P-type metal layer pattern **142**a may be formed in the first region A. The process of etching the P-type metal layer **142** may include a wet etching process. As the wet etching process may be performed, etching damages of the high-k dielectric layer **140** may be decreased compared to performing a dry etching process.

The third mask pattern **132** may be removed. Thereafter, an N-type metal layer **144** may be conformally formed over entire surface of the substrate **100**.

The N-type metal layer **144** may include a metal having an N-work function. In example embodiments, the N-type metal layer **144** may include at least one of lanthanum (La), lanthanum oxide (LaO), tantalum (Ta), tantalum nitride (TaN), niobium (Nb), and titanium nitride (TiN). For example, the N-type metal layer **144** may include a stacked structure including a lanthanum layer and a titanium nitride layer or a stacked structure including a lanthanum oxide layer and a titanium nitride layer.

FIGS. **13** and **14** are enlarged views of portion "F" of FIG. **12**.

Referring to FIG. **13**, the bottom of the first recessed portion **110**a may be substantially flat. In this case, after forming the preliminary P-type metal layer pattern **142**a and the N-type metal layer **144**, the first recessed portion **110**a may have a remaining inner space.

Referring to FIG. **14**, the bottom of the first recessed portion **110**a may have a slope with respect to the flat upper surface of the substrate **100**. In this case, after forming the preliminary P-type metal layer pattern **142**a and the N-type metal layer **144**, the first recessed portion **110**a may have a remaining inner space.

Referring back to FIG. **12**, after forming the N-type metal layer **144**, the second recessed portion **110**b may have remaining inner space. In example embodiments, a width of the inner space of each of the first and second recessed portions **110**a and **110**b may be greater than a sum of thicknesses of the layers formed on the upper sidewall of the first trench **90**a exposed by the first recessed portion **110**a. For example, referring to FIGS. **13** and **14**, the width of the inner space of each of the first and second recessed portions **110**a and **110**b may be greater than a sum of thicknesses of the first interface insulation layer **138**, the high-k dielectric layer **140**, the preliminary P-type metal layer pattern **142**a, and the N-type metal layer **144**.

Referring to FIG. **15**, a lower electrode layer **146**, a barrier layer **148**, an upper electrode layer **150**, and a capping layer may be sequentially stacked on the N-type metal layer **144**. The capping layer may be patterned to form the capping layer pattern **152**.

The lower electrode layer **146** may be formed of a polysilicon layer doped with impurities. Particularly, for forming the lower electrode layer **146**, the polysilicon layer may be deposited on the N-type metal layer **144**. Thereafter, the polysilicon layer positioned on the first region A may be doped with P-type impurities, and the polysilicon layer positioned on the second region B may be doped with N-type impurities. In example embodiments, the lower electrode layer **146** may be formed to fill the first and second recessed portions **110**a and **110**b.

The barrier layer **148** may be formed of, e.g., a titanium nitride layer, a tungsten nitride layer, or a tantalum nitride layer. The upper electrode layer **150** may include a metal material, e.g., tungsten. The capping layer pattern **152** may include, e.g., silicon nitride.

Referring to FIG. **16**, the upper electrode layer **150**, the barrier layer **148**, the lower electrode layer **146**, the N-type metal layer **144**, a preliminary P-type metal layer pattern **142***a*, the high-k dielectric layer **140**, and the first and second interface insulation layers may be sequentially etched using the capping layer pattern **152** as an etching mask to form the first gate structure **160** in the first region A and the second gate structure **162** in the second region B. The etching process may include an anisotropic etching process.

The first gate structure **160** may include a first interface insulation pattern, the first high-k dielectric pattern **140***a*, the first P-type metal pattern **143**, the first N-type metal pattern **144***a*, the first lower electrode **146***a*, the first barrier pattern **148***a*, the first upper electrode **150***a*, and the first capping layer pattern **152***a* stacked. The second gate structure **162** may include a second interface insulation pattern, the second high-k dielectric pattern **140***b*, the second N-type metal pattern **144***b*, the second lower electrode **146***b*, the second barrier pattern **148***b*, the second upper electrode **150***b*, and the second capping layer pattern **152***b* stacked.

In the etching process, it is required that both the preliminary P-type metal layer pattern **142***a* and the N-type metal layer **144** formed in the first and second recessed portions **110***a* and **110***b* are completely removed. If the inner widths of the first and second recessed portions **110***a* and **110***b* were to be narrow, etching gases would not have easily flown into the first and second recessed portions **110***a* and **110***b*, thereby causing the preliminary P-type metal layer pattern **142***a* and the N-type metal layer **144** to partially remain in the first and second recessed portions **110***a* and **110***b*, which in turn, would have generated defects by residues of the preliminary P-type metal layer pattern **142***a* and the N-type metal layer **144**.

In contrast, according to example embodiments, after forming the preliminary P-type metal layer pattern **142***a* and the N-type metal layer **144**, each of the first and second recessed portions **110***a* and **110***b* may maintain sufficient inner space, e.g., due to their larger width. Therefore, the etching gases for etching the metal materials may easily and sufficiently flow into the first and second recessed portions **110***a* and **110***b*, so that both the preliminary P-type metal layer pattern **142***a* and the N-type metal layer **144** positioned in the first and second recessed portions **110***a* and **110***b* may be completely removed.

Further, if the inner widths of the first and second recessed portions **110***a* and **110***b* were to be narrow, the bottom of the first and second recessed portions **110***a* and **110***b* would have had a rounded shape. In this case, the metal layer would have been formed to have a relatively thick thickness in lower portions having the round portions of the first and second recessed portions **110***a* and **110***b*, so that the metal layer would have filled the lower portions of the first and second recessed portions **110***a* and **110***b*. A thickness in the vertical direction of the metal layer formed on the bottoms of the first and second recessed portion would have increased, thereby making removal of the metal layer from the bottoms of the first and second recessed portions **110***a* and **110***b* difficult.

In contrast, according to example embodiments, as the bottoms of the first and second recessed portions **110***a* and **110***b* have flat surfaces, the N-type metal layer **144** and the preliminary P-type metal layer pattern **142***a* may not be folded at the bottoms of the first and second recessed portions **110***a* and **110***b*. The N-type metal layer **144** and the preliminary P-type metal layer pattern **142***a* formed on the bottom of the first and second recessed portions **110***a* and **110***b* may have flat surfaces. Thus, metal layers formed on the bottom of the first and second recessed portions **110***a* and **110***b* may be easily removed.

Referring to FIG. **1** again, the first spacer **164** may be formed on a sidewall of the first gate structure **160**, and the second spacer **166** may be formed on a sidewall of the second gate structure **162**. P-type impurities may be implanted onto the substrate **100** adjacent to both sides of the first gate structure **160** to form the first impurity regions **170**. N-type impurities may be implanted onto the substrate **100** adjacent to both sides of the second gate structure **162** to form the second impurity regions **172**.

As described above, a semiconductor device having high reliability may be manufactured.

FIG. **17** is a cross-sectional view of a semiconductor device in accordance with example embodiments.

Referring to FIG. **17**, the substrate **100** may include a cell array region CA, a first peripheral region A**1**, a second peripheral region A**2**, a third peripheral region B**1**, and a fourth peripheral region B**2**. The first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2** may be disposed around the cell array region CA. Peripheral circuits for driving cells formed in the cell array region CA may be formed in the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2**.

A low voltage PMOS transistor may be formed in the first peripheral region A**1**. A high voltage PMOS transistor may be formed in the second peripheral region A**2**. A low voltage NMOS transistor may be formed in the third peripheral region B**1**. A high voltage NMOS transistor may be formed in the fourth peripheral region B**2**.

Memory cells may be formed in the cell array region CA. For example, the memory cells may include dynamic random-access memory (DRAM) cells.

A cell isolation structure **201** may be formed in a cell trench **92** at the substrate **100** of the cell array region CA. A portion of the substrate **100** of the cell array region CA, in which the cell isolation structure **201** is not formed, may be defined as cell active regions. Each of the cell active regions may have an isolated shape. In a plan view, each of the cell active regions may have an isolated bar shape. Each of the cell active regions may be arranged in a direction oblique to an extending direction of a word line. The substrate **100** may be a single crystal silicon substrate or an SOI substrate.

In the cell array region CA, a width of the cell trench **92** may vary according to a space between adjacent cell active regions, and a stacked structure of the cell isolation structure **201** filling the cell trench may also vary according to the width of the cell trench **92**. In example embodiments, when the width of the cell trench **92** is narrow, the cell trench **92** may be completely filled with an inner wall oxide layer including silicon oxide. In this case, the cell isolation structure **201** may include only silicon oxide, in a cross sectional view.

In some example embodiments, the cell trench may be completely filled with an inner wall oxide layer and a nitride liner, according to the width of the cell trench. In this case, the cell isolation structure may include silicon oxide and silicon nitride, in a cross sectional view.

A buried word line may be formed in the cell active region and the cell isolation structure **201**. The buried word line may extend in a first direction. Impurity regions may be formed at the substrate **100** adjacent to both sides of the word line in the cell active region.

A buffer layer **216** may be formed on the substrate **100** in the cell array region CA. In example embodiments, the buffer layer **216** may include a first insulation layer pattern **210**, a second insulation layer pattern **212**, and a third insulation layer pattern **214** sequentially stacked. The second insulation layer pattern **212** may include a material having a high etch selectivity with respect to the first and third insulation layer patterns **210** and **214**. For example, the second insulation layer pattern **212** may be formed of a silicon nitride layer, and each of the first and third insulation layer patterns **210** and **214** may be formed of a silicon oxide layer. In some example embodiments, the buffer layer **216** may have a two-layer structure including a silicon oxide layer and a silicon nitride layer.

A bit line structure **280** may be formed on the buffer layer **216**. The bit line structure **280** may extend in a second direction perpendicular to the first direction. The bit line structure **280** may include a lower electrode, an upper electrode, a barrier pattern, and a capping layer pattern sequentially stacked. In addition, a portion of the bit line structure **280** may contact the impurity region of the substrate **100**.

The bit line structure **280** may be formed by the same deposition processes as the deposition processes for forming the first to fourth gate structures **260***a*, **260***b*, **260***c*, and **260***d* in the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2**, as will be described in detail below. Thus, the bit line structure **280** may include materials included in the first to fourth gate structures **260***a*, **260***b*, **260***c*, and **260***d*.

The lower electrode may include polysilicon doped with impurities. The upper electrode may include a metal, e.g., tungsten. The capping layer pattern may include silicon nitride.

A bit line spacer **282** may be formed on a sidewall of the bit line structure **280**. An insulation layer may be formed to fill a space between adjacent bit line structures **280**. The insulation layer may, e.g., silicon oxide.

Contact plugs **284** may pass through the insulation layer, and the contact plugs **284** may be formed between adjacent bit line structures **280**. The contact plugs **284** may contact the surface of the substrate **100**. In example embodiments, the contact plugs **284** may include a polysilicon pattern and a metal pattern stacked.

An upper insulation pattern **286** may be formed between the upper portions of the contact plugs **284**. Upper portions of the contact plugs **284** may be electrically isolated by the upper insulation pattern **286**.

A capacitor **290** may be formed on each of the contact plugs **284**. The capacitor **290** may include a lower electrode **290***a*, a dielectric layer **290***b*, and an upper electrode **290***c* stacked. In the capacitor **290**, the lower electrode **290***a* may have a cylindrical shape or a pillar shape.

In the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2**, isolation structures **208***a* may be formed in trenches in the substrate **100**, respectively. In the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2**, a region of the substrate **100** between the isolation structures **208***a* may be defined as a peripheral active region.

The channel layer **130** may be formed on the substrate **100** of the first peripheral region A**1**. The channel layer **130** may be formed of, e.g., silicon germanium. A first gate structure **260***a* may be formed on the channel layer **130**. First impurity regions **264***a* may be formed at the channel layer **130** adjacent to both sides of the first gate structure **260***a*. The first impurity regions **264***a* may be doped with P-type impurities.

A second gate structure **260***b* may be formed on the substrate **100** of the second peripheral region A**2**. Second impurity regions **264***b* may be formed at the substrate **100** adjacent to both sides of the second gate structure **260***b*. The second impurity regions **264***b* may be doped with P-type impurities.

A third gate structure **260***c* may be formed on the third peripheral region B**1**. Third impurity regions **264***c* may be formed at the substrate **100** adjacent to both sides of the third gate structure **260***c*. The third impurity region **264***c* may be doped with N-type impurities.

A fourth gate structure **260***d* may be formed on the fourth peripheral region B**2**. Fourth impurity regions **264***d* may be formed at the substrate **100** adjacent to both sides of the fourth gate structure **260***d*. The fourth impurity regions **264***d* may be doped with N-type impurities.

In example embodiments, each of the isolation structures **208***a* formed in the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2** may be substantially the same as the first and second isolation structures **108***a* and **1098***b* described with reference to FIG. **1**. Each of the isolation structures **208***a* formed in the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2** may include an inner wall oxide pattern **202***a*, a nitride liner **204***a*, and a filling insulation pattern **206***a*.

In the isolation structure **208***a*, the nitride liner **204***a* may protrude from upper portions of the inner wall oxide pattern **202***a* and the filling insulation pattern **206***a*. That is, an uppermost surface of the nitride liner **204***a* may be higher than uppermost surfaces of the inner wall oxide pattern **202***a* and the filling insulation pattern **206***a*. A recessed portion **222** may be formed between the nitride liner **204***a* and the substrate **100** adjacent to the nitride liner **204***a*.

In example embodiments, an inner width of the recessed portion **222** may be greater than a height from a bottom of the recessed portion **222** to an uppermost surface of the nitride liner **204***a*. In example embodiments, the bottom of the recessed portion **222** may have a flat surface. That is, the central portion of the bottom of the recessed portion **222** may not have a rounded shape, and may have a flat surface.

In example embodiments, in the isolation structures **208***a* in the first to fourth peripheral regions A**1**, A**2**, B**1** and B**2**, heights (i.e., vertical levels) of the uppermost surfaces of the nitride liners **204***a* may be substantially the same. In example embodiments, in each of the isolation structures **208***a* of the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2**, the uppermost surface of each of the nitride liners **204***a* may be substantially coplanar with an upper surface of the substrate **100** adjacent to the nitride liner **204***a*.

The first gate structure **260***a* may include a first interface insulation pattern **224***a*, a first high-k dielectric pattern **230***a*, a first P-type metal pattern **232***a*, a first N-type metal pattern **234***a*, a first lower electrode **250***a*, a first barrier pattern, a first upper electrode **254***a*, and a first capping layer pattern **256***a* sequentially stacked. The second gate structure **260***b* may include a second interface insulation pattern **226***a*, a second high-k dielectric pattern **230***b*, a second P-type metal pattern **232***b*, a second N-type metal pattern **234***b*, a second lower electrode **250***b*, a second barrier pattern, a second upper electrode **254***b*, and a second capping layer pattern **256***b* sequentially stacked.

A gate length of the first gate structure **260***a* may be less than a gate length of the second gate structure **260***b*. The gate length of the first and second gate structures **260***a* and **160***b* may be a line width of the gate structure.

A thickness of the first interface insulation pattern **224***a* may be less than a thickness of the second interface insulation pattern **226***a*. In example embodiments, a material of the first interface insulation pattern **224***a* may be different from a material of the second interface insulation pattern **226***a*. For example, the first interface insulation pattern **224***a* may include silicon oxide, and the second interface insulation pattern **226***a* may include silicon oxynitride.

The first and second gate structures **260***a* and **260***b* may have substantially the same stacked structure, except for the first and second interface insulation patterns **224***a* and **226***a*. That is, in the first and second gate structures **260***a* and **260***b*, a stacked structure formed on the first interface insulation pattern **224***a* and a stacked structure formed on the second interface insulation pattern **226***a* may be substantially the same. In the first and second gate structures **260***a* and **260***b*, the stacked structures formed on the first and second interface insulation patterns **224***a* and **226***a* may be substantially the same as the stacked structure formed on the first interface insulation pattern in the first gate structure described with reference to FIG. **1**.

The third gate structure **260***c* may include a third interface insulation pattern **224***b*, a third high-k dielectric pattern **230***c*, a third N-type metal pattern **234***c*, a third lower electrode **250***c*, a third barrier pattern, a third upper electrode **254***c*, and a third capping layer pattern **256***c* sequentially stacked. The fourth gate structure **260***d* may include a fourth interface insulation pattern **226***b*, a fourth high-k dielectric pattern **230***d*, a fourth N-type metal pattern **234***d*, a fourth lower electrode **250***d*, a fourth barrier pattern, a fourth upper electrode **254***d*, and a fourth capping layer pattern **256***d* sequentially stacked.

A gate length of the third gate structure **260***c* may be less than a gate length of the fourth gate structure **260***d*. A thickness of the third interface insulation pattern **224***b* may be less than a thickness of the fourth interface insulation pattern **226***b*. In example embodiments, a material of the third interface insulation pattern **224***b* may be different from a material of the fourth interface insulation pattern **226***b*. For example, the third interface insulation pattern **224***b* may include silicon oxide, and the fourth interface insulation pattern **226***b* may include silicon oxynitride.

The third and fourth gate structures **260***c* and **260***d* may have the same stacked structure, except for the third and fourth interface insulation patterns **224***b* and **226***b*. That is, in the third and fourth gate structures **260***c* and **260***d*, a stacked structure formed on the third interface insulation pattern **224***b* and a stacked structure formed on the fourth interface insulation pattern **226***b* may be substantially the same. In the third and fourth gate structures **260***c* and **260***d*, the stacked structures formed on the third and fourth interface insulation patterns **224***b* and **226***b* may be substantially the same as the stacked structure formed on the second interface insulation pattern in the second gate structure described with reference to FIG. **1**.

As described above, the bit line structure **280** in the cell array region CA may include a first structure in which the lower electrode, the barrier pattern, the upper electrode, and the capping layer pattern are stacked. In addition, in the first and second gate structures **260***a* and **260***b*, the first structure including the lower electrode, the barrier pattern, the upper electrode, and the capping layer pattern may be stacked on each of the first and second N-type metal patterns **234***a* and **234***b*. In the third and fourth gate structures **260***c* and **260***c*, the first structure including the lower electrode, the barrier pattern, the upper electrode, and the capping layer pattern may be stacked on each of the third and fourth N-type metal patterns **234***c* and **234***d*.

First to fourth spacers **262***a*, **262***b*, **262***c* and **262***d* may be formed on sidewalls of the first to fourth gate structures **260***a*, **260***b*, **260***c* and **260***d*, respectively. An insulating interlayer **266** may be formed to cover the first to fourth gate structures **260***a*, **260***b*, **260***c* and **260***d*.

An upper capping layer **270** may be formed on the insulating interlayer **266** in the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2**.

As described above, the DRAM cells may be formed in the cell array region CA. The low voltage PMOS transistor, the high voltage PMOS transistor, the low voltage NMOS transistor, and the high voltage NMOS transistors including high-k dielectric layers may be formed in the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2**. At least the low voltage PMOS transistor may be formed on the channel layer **130** including the silicon germanium. The isolation structure including the nitride liner **204***a* having the protruding portion may be formed in the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2**. The uppermost surface of the nitride liner **204***a* may be coplanar with the upper surface of the substrate **100** adjacent thereto.

FIG. **18** is a cross-sectional view of a semiconductor device in accordance with example embodiments. The semiconductor device in FIG. **18** may be substantially the same as the semiconductor device shown in FIG. **17**, except for further including a second channel layer.

Referring to FIG. **18**, a second channel layer **130***a* may be further formed on the substrate **100** of the second peripheral region A**2**. The second channel layer **130***a* may be formed of, e.g., silicon germanium. The second gate structure **260***b* may be formed on the second channel layer **130***a*. That is, the high voltage PMOS transistor may be formed on the second channel layer **130***a*.

FIGS. **19** to **26** are cross-sectional views of stages in a method of manufacturing a semiconductor device in accordance with example embodiments.

Referring to FIG. **19**, the substrate **100** may include the cell array region CA, the first peripheral region A**1**, the second peripheral region A**2**, the third peripheral region B**1**, and the fourth peripheral region B**2**. A portion of the substrate **100** may be etched to form the cell trench **92** and the peripheral trench **90**. The cell trench **92** may have a width less than a width of the peripheral trench **90**.

A preliminary isolation structure **207***a* including a preliminary inner wall oxide pattern **201***a*, a nitride liner **204***a*, and a preliminary filling insulation pattern **205***a* may be formed in the peripheral trench **90**. In example embodiments, the cell trench **92** may be filled with the inner wall oxide pattern, and thus a cell isolation structure **201** including the inner wall oxide pattern may be formed. In some example embodiments, a portion of the cell trench **92** may be filled with the inner wall oxide pattern and the nitride liner.

Word lines buried in the substrate **100** may be formed in the cell array region CA. An ion implantation process may be performed to form impurity regions in the substrate **100** adjacent to both sides of the word line in the cell array region CA.

Referring to FIG. **20**, a first insulation layer, a second insulation layer, and a third insulation layer may be sequentially stacked on an entire surface of the substrate **100** and the cell isolation structure **201**. The first insulation layer, the second insulation layer, and the third insulation layer may be patterned to form a buffer layer **216** including a first insulation layer pattern **210**, a second insulation layer pattern **212** and a third insulation layer pattern **214** in the cell array region CA.

A first mask pattern **218** may be formed on the substrate **100**. The first mask pattern **218** may cover the cell array region CA and the second to fourth peripheral regions A2, B1, and B2, and may expose the first peripheral region A1. The first mask pattern **218** may be formed of, e.g., a silicon oxide layer.

When the first mask pattern **218** is formed by an etching process, upper portions of a preliminary inner wall oxide pattern **201***a* and a preliminary filling insulation pattern in the preliminary isolation structure **207***a* in the first peripheral region A1 may be only etched by over etching in the etching process. Thus, etched thicknesses of the preliminary inner wall oxide pattern **201***a* and the preliminary filling insulation pattern **205***a* may be decreased.

Therefore, the isolation structure **208***a* including an inner wall oxide pattern **202***a*, the nitride liner **204***a*, and a filling insulation pattern **206***a* may be formed in the first peripheral region A1. The nitride liner **204***a* may protrude from upper surfaces of the inner wall oxide pattern **202***a* and the filling insulation pattern **206***a*.

The channel layer **130** may be formed on the substrate **100** of the first peripheral region A1 exposed by the first mask pattern **218**. The channel layer **130** may be formed of, e.g., a silicon germanium layer. The channel layer **130** may be formed by a selective epitaxial growth (SEG) process.

In some example embodiments, the first mask pattern may expose the first and second peripheral regions A1 and A2. In this case, the channel layer may be formed on the substrate **100** of the first peripheral region A1 and the second peripheral region A2. Therefore, the semiconductor device shown in FIG. **18** may be manufactured by subsequent processes.

Referring to FIG. **21**, a photoresist pattern **220** may be formed to cover the channel layer **130** and the isolation structure **208***a* in the first peripheral region A1. Thereafter, the first mask pattern **218** may be removed.

When the first mask pattern **218** is removed by an etching process, upper portions of the preliminary inner wall oxide pattern **201***a* and the preliminary filling insulation pattern **205***a* in the preliminary isolation structure **207***a* in the second to fourth peripheral regions A2, B1, and B2 may be only etched by overetching in the etching process. Thus, etched thicknesses of the preliminary inner wall oxide pattern **201***a* and the preliminary filling insulation pattern **205***a* may be decreased. Therefore, the isolation structure **208***a* including the inner wall oxide pattern **202***a*, the nitride liner **204***a*, and the filling insulation pattern **206***a* may be formed in each of the second to fourth peripheral regions A2, B1, and B2.

In the first to fourth peripheral regions A1, A2, B1, and B2, each of the isolation structures **208***a* may include the inner wall oxide pattern **202***a* and the nitride liner **204***a* protruding from the filling insulation pattern **206***a*. An uppermost surface of the inner wall oxide pattern **202***a* may be lower than an uppermost surface of the nitride liner **204***a*. Thus, the recessed portion **222** may be formed between the nitride liner **204***a* and the substrate **100** adjacent the nitride liner **204***a*.

In the isolation structures **208***a* formed in the first to fourth peripheral regions A1, A2, B1, and B2, heights of upper surfaces of the nitride liners **204***a* may be substantially the same. In example embodiments, in the isolation structures **208***a* in the first to fourth peripheral regions A1, A2, B1, and B2, the uppermost surfaces of the nitride liners **204***a* may be substantially coplanar with an uppermost surface of the substrate **100** adjacent thereto.

In example embodiments, in the isolation structures **208***a* in the first to fourth peripheral regions A1, A2, B1, and B2, vertical levels of bottoms of the recessed portions **222** may be substantially the same.

FIGS. **22** and **23**, a high voltage interface insulation layer **226** may be formed on the substrate **100** of the second and fourth peripheral regions A2 and B2. The high voltage interface insulation layer **226** may be formed of, e.g., silicon oxynitride.

A low voltage interface insulation layer **224** may be formed on the substrate **100** and the channel layer **130** in the first and third peripheral regions A1 and B1. The low voltage interface insulation layer **224** may be formed of, e.g., silicon oxide. The low voltage interface insulation layer **224** may be formed to have a thickness less than a thickness of the high voltage interface insulation layer **226**.

A high-k dielectric layer **230** may be conformally formed over an entire surface of the substrate **100**. The high-k dielectric layer **230** may be formed by a deposition process, e.g., chemical vapor deposition (CVD) or atomic layer deposition (ALD). A P-type metal layer **232** may be conformally formed on the high-k dielectric layer **230**.

A second mask pattern may be formed on the P-type metal layer **232**. The second mask pattern may cover the first and second peripheral regions A1 and A2, and the second mask pattern may expose the third and fourth peripheral regions B1 and B2 and the cell array region CA. The P-type metal layer in the cell array region CA and the third and fourth peripheral regions B1 and B2 may be removed using the second mask pattern as an etching mask. Thus, the high-k dielectric layer **230** may be exposed on the cell array region CA and the third and fourth peripheral regions B1 and B2. The removing process may include wet etching. An N-type metal layer **234** may be formed on the P-type metal layer **232** and the high-k dielectric layer **230**.

FIG. **23** is an enlarged view of a portion "G" of FIG. **22**.

Referring to FIG. **23**, the high-k dielectric layer **230**, the P-type metal layer **232**, and the N-type metal layer **234** may be stacked on the first and second peripheral regions A1 and A2. The high-k dielectric layer **230** and the N-type metal layer **234** may be stacked on the third and fourth peripheral regions B1 and B2.

In FIG. **22**, in order to simplify the drawing, the high-k dielectric layer **230**, the P-type metal layer **232**, and the N-type metal layer **234** in the first and second peripheral regions A1 and A2 may be illustrated as a first layer **240**, and the high-k dielectric layer **230** and the N-type metal layer **234** in the third and fourth peripheral regions B1 and B2 and the cell array region CA may be illustrated as a second layer **242**. After forming the first and second layers, recessed portions **222** of the isolation structure in the first to fourth peripheral regions A1, A2, B1 and B2 may have remaining inner spaces.

Referring to FIG. **24**, a third mask pattern may be formed to cover the N-type metal layer **234** in the first to fourth peripheral regions A1, A2, B1, and B2. The third mask pattern may expose the cell array region CA. The N-type metal layer **234** and the high-k dielectric layer **230** in the cell array region CA may be removed using the third mask pattern. Then, the third mask pattern may be removed.

A lower electrode layer **250** may be formed over the entire surface of the substrate **100**. Portions of the lower electrode layer **250**, the buffer layer **216**, and the substrate **100** in the cell array region may be etched to form a first opening. The first opening may correspond to a position for forming a bit line contact. In an etching process for forming the first opening, an upper portion of the cell isolation structure **201** may be removed together.

A polysilicon layer may be formed to fill the first opening, and the polysilicon layer may be planarized by a planarization process. Thus, a polysilicon pattern **252** may be formed in the first opening.

A barrier layer and an upper electrode layer **254** may be formed on the lower electrode layer **250** and the polysilicon pattern **252**. A capping layer pattern **256** may be formed on the upper electrode layer **254**. The capping layer pattern **256** may cover entire of the cell array region CA. In addition, the capping layer pattern **256** may be also disposed at portions for forming the first to fourth gate structures in the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2**.

Referring to FIG. **25**, the upper electrode layer **254**, the barrier layer, the lower electrode layer **250**, the P-type metal layer **232**, the N-type metal layer **234**, the high-k dielectric layer **230**, the high voltage interface insulation layer **226**, and the low voltage interface insulation layer **224** in the first to fourth peripheral regions A**1**, A**2**, B**1** and B**2** may be etched using the capping layer pattern **256** as an etching mask to form the first to fourth gate structures **260***a*, **260***b*, **260***c* and **260***d*. First to fourth spacers **262***a*, **262***b*, **262***c*, and **262***d* may be formed on sidewalls of the first to fourth gate structures **260***a*, **260***b*, **260***c*, and **260***d*, respectively.

P-type impurities may be doped onto the substrate adjacent to the sidewalls of the first and second gate structures **260***a* and **260***b* to form first and second impurity regions **264***a* and **264***b*, respectively. N-type impurities may be doped onto the substrate adjacent to the sidewalls of the third and fourth gate structures **260***c* and **260***d* to form third and fourth impurity regions **264***c* and **264***d*, respectively.

An insulating interlayer **266** may be formed on the entire surface of the substrate **100** to cover the first to fourth gate structures **260***a*, **260***b*, **260***c*, and **260***d*. An upper portion of the insulating interlayer **266** may be planarized. An upper capping layer **270** may be formed on the capping layer pattern **256** in the cell array region CA and the insulating interlayer **266** in the first to fourth peripheral regions A**1**, A**2**, B**1**, and B**2**.

Referring to FIG. **26**, the upper capping layer **270**, the capping layer pattern **256**, the upper electrode layer **254**, the barrier layer, the lower electrode layer **250**, and the polysilicon pattern **252** in the cell array region CA may be etched using a mask to form a bit line structure **280**. A bit line spacer **282** may be formed on a sidewall of the bit line structure **280**.

An insulation layer may be formed to fill a gap between bit line structures **280**. Portions of the insulation layer and the buffer layer **216** may be etched to form a second opening exposing the surface of the substrate **100**. A contact plug **284** may be formed to fill the second opening. An upper insulation pattern **286** may be formed between upper portions of adjacent contact plugs **284**.

Referring to FIG. **17** again, a capacitor **290** may be formed on the contact plug **184**.

As described above, the DRAM cells may be formed in the cell array region, and the low voltage PMOS transistor, the high voltage PMOS transistor, the low voltage NMOS transistor, and the high voltage NMOS transistor including the high-k dielectric layer may be formed in the first to fourth peripheral regions. At least the low voltage PMOS transistor may be formed on the channel layer including the silicon germanium.

By way of summation and review, example embodiments provide a semiconductor device having good electrical characteristics and high reliability. That is, in the semiconductor device, in accordance with example embodiments, an isolation structure may include a nitride liner having a protruding portion that defines a recessed portion with an increased width. A metal material may not remain in the recessed portion, thereby decreasing defects, e.g., due to residual metal materials in the recessed portion, in the semiconductor device.

Example embodiments have been disclosed herein, and although specific terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent to one of ordinary skill in the art as of the filing of the present application, features, characteristics, and/or elements described in connection with a particular embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of skill in the art that various changes in form and details may be made without departing from the spirit and scope of the present invention as set forth in the following claims.

What is claimed is:

1. A semiconductor device, comprising:

a substrate including a first active region, a second active region, and an isolation structure between the first active region and the second active region;

a first gate structure arranged on the first active region of the substrate and including a first high-k dielectric pattern and a first metal pattern; and

a second gate structure arranged on the second active region of the substrate and including a second high-k dielectric pattern and a second metal pattern,

wherein the isolation structure includes an inner wall oxide pattern, a liner on the inner wall oxide pattern, and a filling insulation pattern on the liner,

wherein the inner wall oxide pattern of the isolation structure includes a vertical portion and a horizontal portion,

wherein the liner of the isolation structure includes a vertical portion and a horizontal portion,

wherein a thickness of the vertical portion of the inner wall oxide pattern of the isolation structure in a first direction that is parallel to a top surface of the substrate is greater than a thickness of the vertical portion of the liner of the isolation structure in the first direction,

wherein a top surface of the vertical portion of the liner of the isolation structure is higher than a top surface of the vertical portion of the inner wall oxide pattern of the isolation structure,

wherein the first high-k dielectric pattern of the first gate structure includes hafnium, and

wherein the top surface of the vertical portion of the inner wall oxide pattern is substantially flat without a curved portion.

2. The semiconductor device as claimed in claim **1**,

wherein the second high-k dielectric pattern of the second gate structure includes hafnium.

3. The semiconductor device as claimed in claim **1**,

wherein a material of the first high-k dielectric pattern of the first gate structure includes an oxide including hafnium (Hf) and silicon (Si).

4. The semiconductor device as claimed in claim **1**,

wherein a material of the second high-k dielectric pattern of the second gate structure includes hafnium (Hf), silicon (Si), and aluminum (Al).

5. The semiconductor device as claimed in claim **1**,

wherein each of the first high-k dielectric pattern of the first gate structure and the second high-k dielectric pattern of the second gate structure includes at least one of hafnium oxide (HfO), hafnium silicon oxide (HfSiO), hafnium oxynitride (HfON), hafnium silicon oxynitride (HfSiON), lanthanum oxide (LaO), lanthanum aluminum oxide (LaAlO), zirconium oxide (ZrO), Zirconium silicon oxide (ZrSiO), zirconium oxynitride (ZrON), zirconium silicon oxynitride (ZrSiON), tantalum oxide (TaO), titanium oxide (TiO), barium strontium titanium oxide (BaSrTiO), barium titanium oxide (BaTiO), strontium titanium oxide (SrTiO), yttrium oxide (YO), aluminum oxide (Al2O3), or lead scandium tantalum oxide (PbScTaO).

6. The semiconductor device as claimed in claim **1**, wherein the first metal pattern of the first gate structure includes titanium nitride (TiN).

7. The semiconductor device as claimed in claim **1**, wherein the second metal pattern of the second gate structure includes titanium nitride (TiN).

8. The semiconductor device as claimed in claim **1**, wherein the first metal pattern of the first gate structure includes at least one of lanthanum (La), lanthanum oxide (LaO), tantalum (Ta), tantalum nitride (TaN), niobium (Nb), or titanium nitride (TiN).

9. The semiconductor device as claimed in claim **1**, wherein the second metal pattern of the second gate structure includes at least one of aluminum (Al), aluminum oxide (AlO), titanium nitride (TiN), tungsten nitride (WN), or ruthenium oxide (RuO).

10. The semiconductor device as claim in claim **1**, wherein the first gate structure includes a first interface insulation pattern that is between the first active region of the substrate and the first high-k dielectric pattern of the first gate structure, the first interface insulation pattern of the first gate structure including silicon oxide and/or silicon oxynitride.

11. The semiconductor device as claim in claim **1**, wherein the second gate structure includes a second interface insulation pattern that is between the second active region of the substrate and the second high-k dielectric pattern of the second gate structure, the second interface insulation pattern of the second gate structure including silicon oxide and/or silicon oxynitride.

12. The semiconductor device as claimed in claim **1**, wherein the top surface of the vertical portion of the inner wall oxide pattern of the isolation structure is lower than the top surface of the substrate.

13. The semiconductor device as claimed in claim **1**, wherein the top surface of the vertical portion of the liner of the isolation structure is higher than a top surface of the filling insulation pattern of the isolation structure.

14. The semiconductor device as claimed in claim **1**, wherein a thickness of the horizontal portion of the inner wall oxide pattern of the isolation structure in a second direction that is perpendicular to the first direction is greater than a thickness of the horizontal portion of the liner of the isolation structure in the second direction.

15. The semiconductor device as claimed in claim **1**, wherein each of the inner wall oxide pattern and the filling insulation pattern of the isolation structure includes silicon oxide, and the liner of the isolation structure includes silicon nitride.

16. The semiconductor device as claimed in claim **1**, wherein each of the first gate structure and the second gate structure includes a polysilicon pattern, a third metal pattern, and a capping pattern, the third metal pattern of each of the first gate structure and the second gate structure including titanium nitride (TiN) and tungsten (W).

17. A semiconductor device, comprising:
a substrate including a first active region, a second active region, and a trench between the first active region and the second active region, the trench having a side and a bottom;
an isolation structure in the trench of the substrate;
a first gate structure arranged on the first active region of the substrate and including a first high-k dielectric pattern and a first metal pattern; and
a second gate structure arranged on the second active region of the substrate and including a second high-k dielectric pattern and a second metal pattern,
wherein the isolation structure includes an inner wall oxide pattern that is on the side and the bottom of the trench of the substrate, a liner that is on the inner wall oxide pattern, and a filling insulation pattern that is on the liner,
wherein a top surface of the liner of the isolation structure is higher than a top surface of the inner wall oxide pattern of the isolation structure and a top surface of the filling insulation pattern of the isolation structure,
wherein a width between a side surface of the substrate that is exposed by the trench of the substrate and the top surface of the liner of the isolation structure is greater than a height between the top surface of the inner wall oxide pattern of the isolation structure and the top surface of the liner of the isolation structure,
wherein a material of the first high-k dielectric pattern of the first gate structure includes an oxide including hafnium (Hf) and silicon (Si),
wherein a material of the second high-k dielectric pattern of the second gate structure includes an oxide including hafnium (Hf), silicon (Si) and aluminum (Al),
wherein the inner wall oxide pattern of the isolation structure includes a vertical portion and a horizontal portion, and
wherein the top surface of the vertical portion of the inner wall oxide pattern is substantially flat without a curved portion.

18. The semiconductor device as claimed in claim **17**, wherein:
each of the inner wall oxide pattern and the filling insulation pattern of the isolation structure includes silicon oxide,
the liner of the isolation structure includes silicon nitride, and
each of the first metal pattern of the first gate structure and the second metal pattern of the second gate structure includes titanium nitride (TiN).

19. The semiconductor device as claimed in claim **17**, wherein a thickness of the inner wall oxide pattern of the isolation structure is greater than a thickness of the liner of the isolation structure.

20. A semiconductor device, comprising:
a substrate including a first active region, a second active region, and an isolation structure between the first active region and the second active region;
a first gate structure arranged on the first active region of the substrate and including a first high-k dielectric pattern and a first metal pattern; and
a second gate structure arranged on the second active region of the substrate and including a second high-k dielectric pattern and a second metal pattern,

wherein the isolation structure includes an inner wall oxide pattern, a liner on the inner wall oxide pattern, and a filling insulation pattern on the liner,

wherein a thickness of the inner wall oxide pattern of the isolation structure is greater than a thickness of the liner of the isolation structure,

wherein a top surface of the liner of the isolation structure is higher than a top surface of the inner wall oxide pattern of the isolation structure and a top surface of the filling insulation pattern of the isolation structure,

wherein a material of the first high-k dielectric pattern of the first gate structure includes an oxide including hafnium (Hf) and silicon (Si),

wherein a material of the second high-k dielectric pattern of the second gate structure includes an oxide including hafnium (Hf), silicon (Si) and aluminum (Al),

wherein each of the inner wall oxide pattern and the filling insulation pattern of the isolation structure includes silicon oxide (SiO),

wherein the liner of the isolation structure includes silicon nitride (SiN),

wherein each of the first metal pattern of the first gate structure and the second metal pattern of the second gate structure includes titanium nitride (TiN),

wherein the inner wall oxide pattern of the isolation structure includes a vertical portion and a horizontal portion, and

wherein the top surface of the vertical portion of the inner wall oxide pattern is substantially flat without a curved portion.

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